From critical scaling to baryon number fluctuations: probing the QCD critical point

Shi Yin

Alumni Seminar, 08.10.2026

Based on: 1. Fu, Luo, Pawlowski, Rennecke and **SY**, arXiv:2308.15508
2. Recchi and **SY**, arXiv:2609.17328
3. **SY**, in preparation

With the help of Claude

CRC-TR 211
Strong-interaction matter under extreme conditions

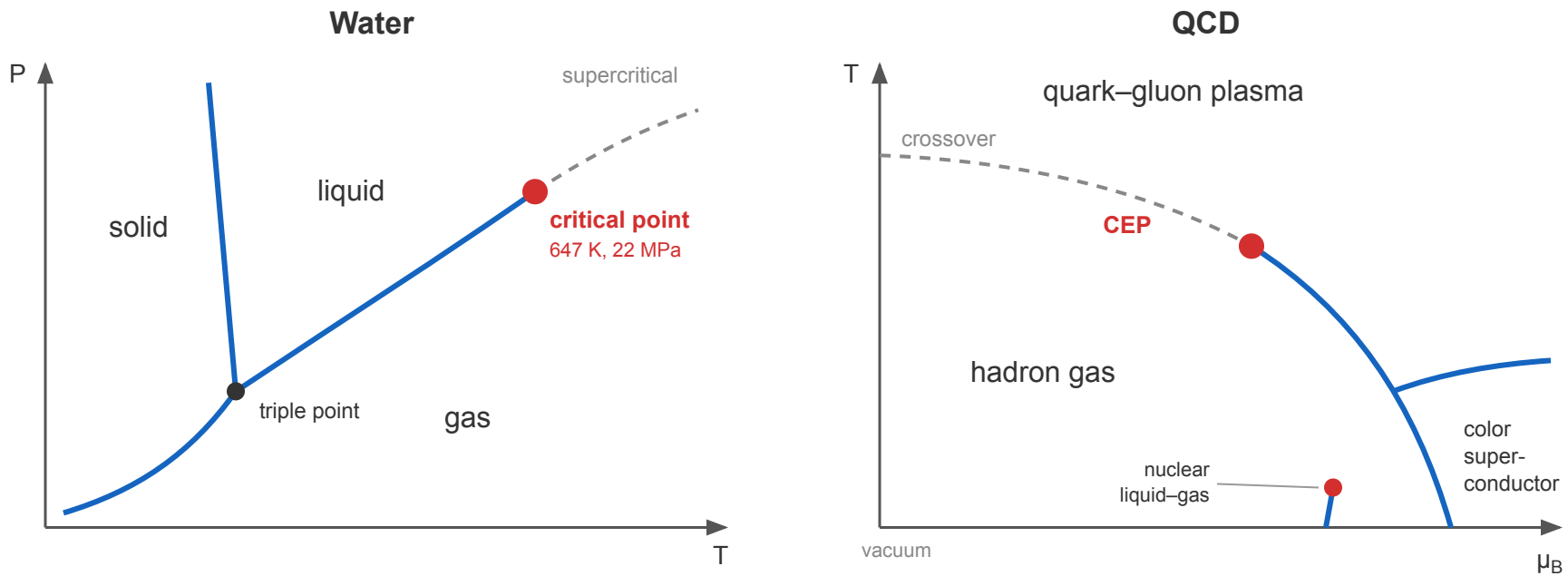
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fQCD

Outline

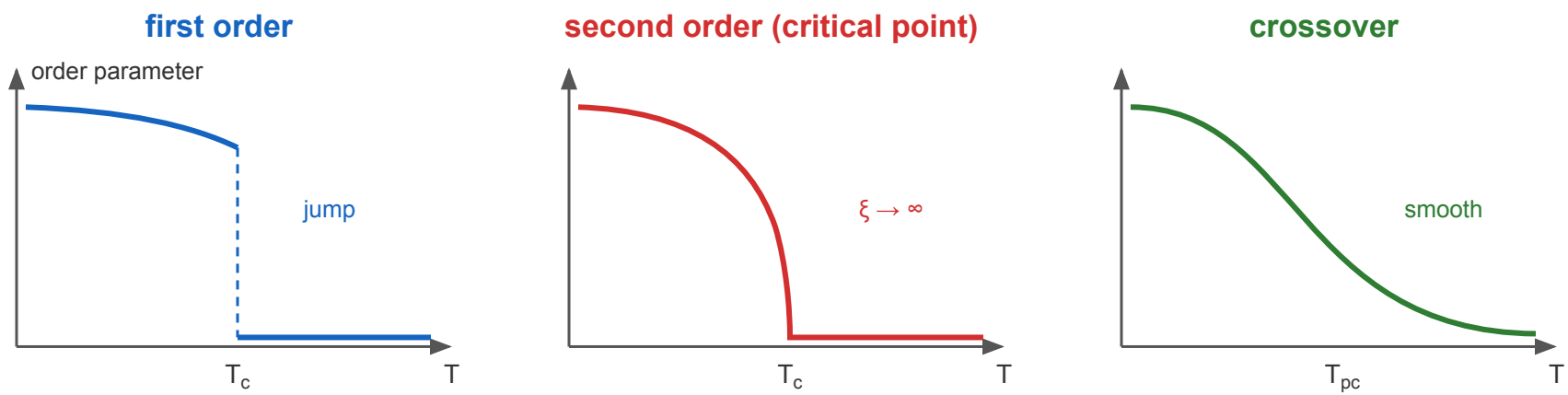
- QCD phase diagram, phase transitions and universality
- Heavy-ion collisions
- Baryon number fluctuations
- QCD-assisted low energy effective theory
- Fluctuations near the liquid–gas transition
- Upgraded fRG computations and critical behavior
- Summary and Outlook

Phases of strongly interacting matter



- Water: the first-order liquid–gas line **ends** at a critical point; beyond it liquid and gas are connected by a crossover
- QCD: hadrons $\leftrightarrow$ quark–gluon plasma, crossover at small μ_B, conjectured first-order line ending at the **CEP**
- Both critical points (and the nuclear liquid–gas one) belong to the **3D Ising** universality class

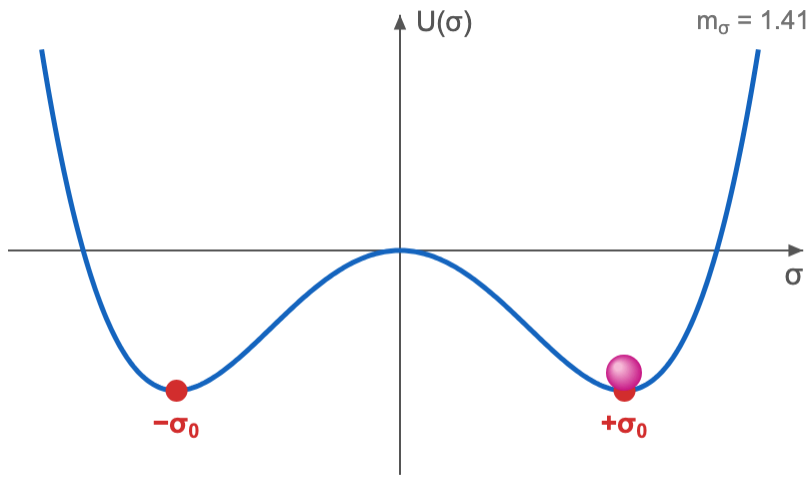
Phase transitions in a nutshell



- Order parameters of QCD: chiral condensate $\langle \bar{q}q \rangle$ (chiral symmetry), Polyakov loop L (confinement)
- Critical point: end point of a first-order line, second-order transition with correlation length $\xi \to \infty$
- Critical behavior is **universal**: it only depends on symmetry and dimension, not on the microscopic details

O(4) vs Z_2: shape of the effective potential

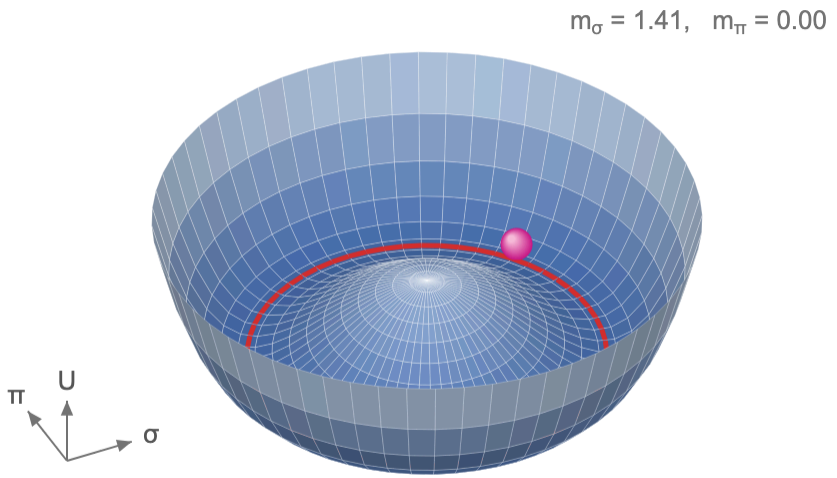
Z_2: $\sigma \to -\sigma$



two isolated vacua $\pm\sigma_0$: a discrete choice

curved in every direction: $m_\sigma^2 = U''(\sigma_0) > 0$

O(4): rotations of (σ, π_1, π_2, π_3)



continuous valley of vacua (S^3, red ring)

rolling along the flat valley is free: $m_\pi = 0$

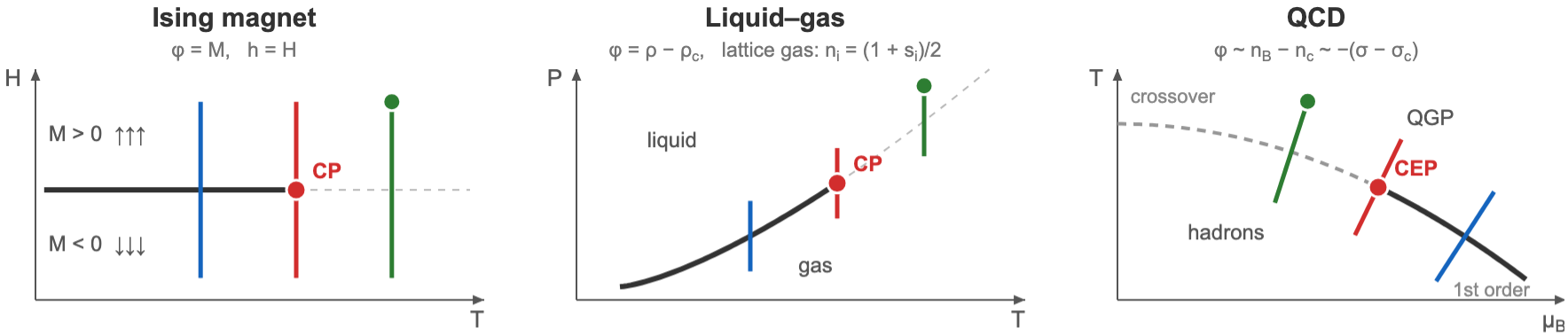
T / T_c 0.50 ☐ explicit breaking $-h\sigma$

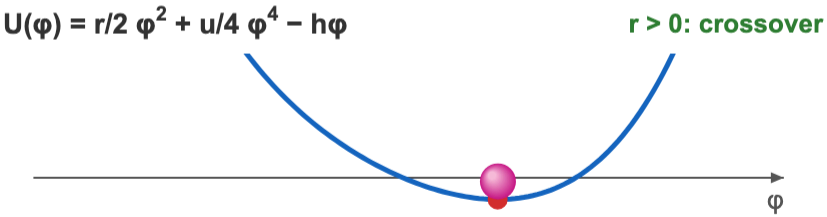
- $U(\sigma) = \frac{a}{2}\sigma^2 + \frac{\lambda}{4}\sigma^4 - h\sigma$, one real field
- two isolated vacua $\pm\sigma_0$: no flat direction, all modes massive
- 3D Ising class: QCD critical endpoint

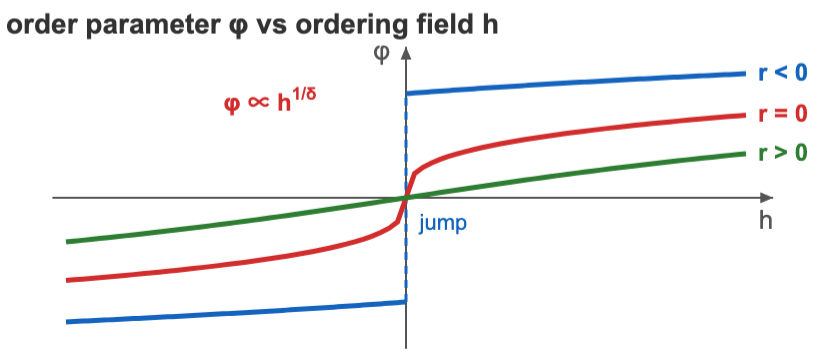
- $U(\rho) - h\sigma$, $\rho = \frac{1}{2}(\sigma^2 + \vec{\pi}^{\,2})$
- valley of vacua S^3: flat directions ⇒ 3 Goldstone pions, $m_\pi^2 = U'(\rho_0) = h/\sigma_0$
- 3D O(4) class: chiral transition ($N_f = 2,\ m_q = 0$)

$h \neq 0$: the pion direction stays curved, $m_\pi^2 = h/\sigma_0 > 0$ ⇒ only the σ direction can turn flat ⇒ a critical point at $m_q \neq 0$ (e.g. the QCD CEP) is Z_2, not O(4)

Why is the QCD CEP an Ising critical point?







▶ r < 0: first-order line ▶ r = 0: critical point ▶ r > 0: crossover

- Only **one** mode turns soft ($\xi \to \infty$): one real field $\phi \Rightarrow U(\phi) = \frac{r}{2}\phi^2 + \frac{u}{4}\phi^4 - h\phi$ (ϕ^3 removed by a shift)
- First-order line $= \{h = 0,\ r < 0\}$: two degenerate minima ($\uparrow$ / $\downarrow$, liquid/gas, hadrons/quark matter)
- Same $d = 3$ and same order parameter ⇒ same exponents $\beta \simeq 0.326,\ \gamma \simeq 1.237,\ \nu \simeq 0.630,\ \delta \simeq 4.79$

In QCD the Z_2 is not a symmetry of the Lagrangian: it **emerges** at the CEP. ϕ mixes σ, n_B and ε; $(T, \mu_B) \to (r, h)$ is a non-universal map

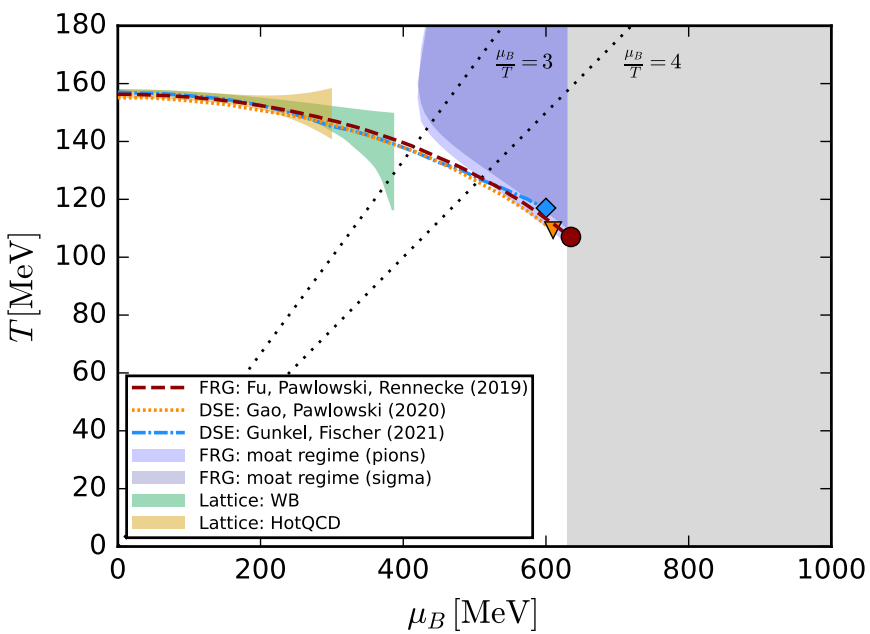
Critical scaling near the CEP

$$\xi \sim |r|^{-\nu}, \qquad \phi \sim (-r)^{\beta}, \qquad \chi \sim |r|^{-\gamma}, \qquad \phi\big|_{r=0} \sim h^{1/\delta}$$

$$\phi = h^{1/\delta} f_G(z), \qquad z = r\, h^{-1/(\beta\delta)}$$

- Close to the CEP all singular behavior follows from **one** universal scaling function $f_G(z)$
- Cumulants $\leftrightarrow$ derivatives of f_G: universal sign pattern of κ_4
- Non-universal: amplitudes and the size of the scaling region

What do we know about the QCD phase diagram?



- $\mu_B = 0$: chiral **crossover** at $T_{pc} \approx 156$ MeV (lattice QCD)
- $\mu_B > 0$: sign problem, lattice results limited to $\mu_B/T \lesssim 2$–3
- Functional QCD (fRG, DSE) estimates of the CEP converge:

$$600\ \mathrm{MeV} \lesssim \mu_{B,\mathrm{CEP}} \lesssim 650\ \mathrm{MeV}$$

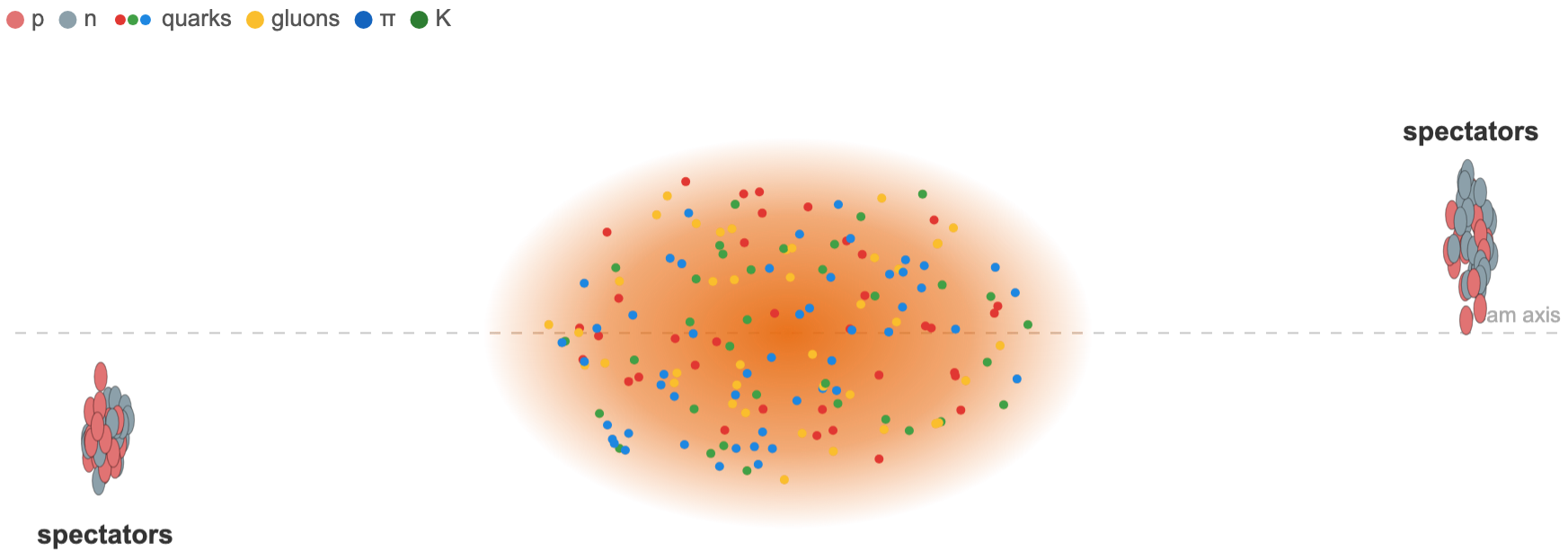
- Where is the CEP? Can experiments see it?

fRG: Fu, Pawlowski, Rennecke (2019); DSE: Gao, Pawlowski (2020); Gunkel, Fischer (2021); lattice: WB, HotQCD

Outline

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- Heavy-ion collisions
- Baryon number fluctuations
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- Summary and Outlook

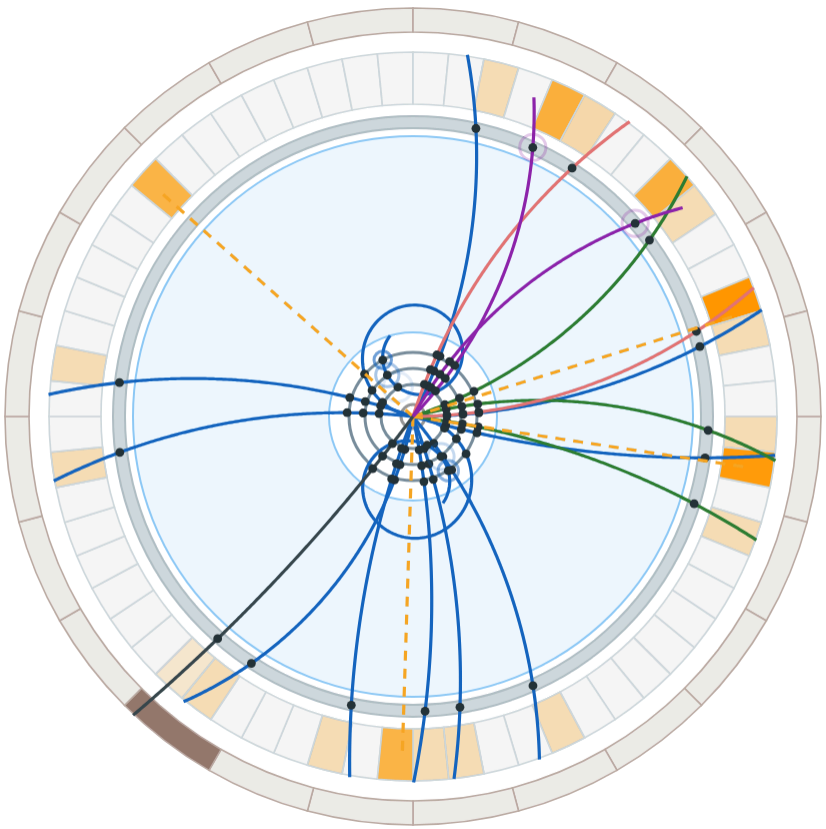
Heavy-ion collisions



Approach | Collision | QGP | Hadronization | Freeze-out

A hot, dense quark–gluon plasma forms and expands (~1–10 fm/c)

Particles entering the detector



Detector layers

- Beam pipe
- Inner tracker (silicon)
- Time Projection Chamber (TPC)
- Time-of-flight (TOF)
- Electromagnetic calorimeter
- Muon chambers

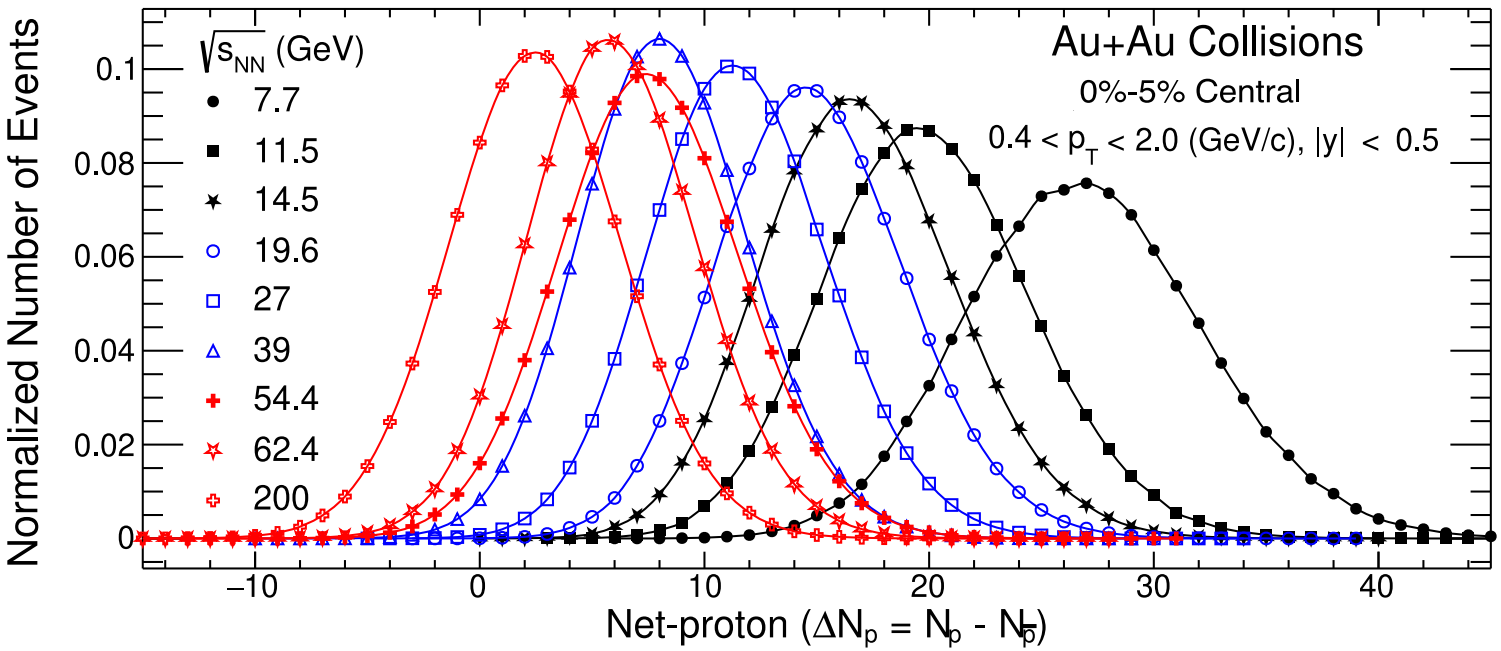
Particles

π±	K±	p
e±	μ±	γ

- Transverse view: beams ⊥ screen
- Magnetic field bends charged tracks, radius ∝ pT / |q|
- Low-pT particles curl up inside the TPC
- Photons leave no track, only energy in the calorimeter
- Muons pass through all layers

Schematic, not to scale

Counting particles in the detector



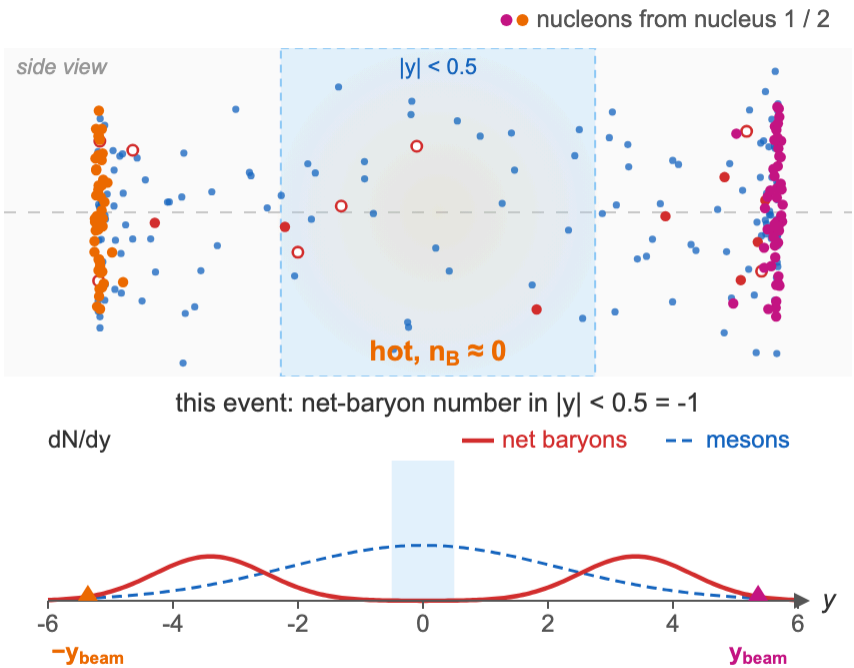
STAR Collaboration, arXiv:2001.02852

- The probability distribution of net-proton
- The distribution changes with the collision energy
- Collision energy is related to the density

High vs low collision energy

Same Au+Au, different $\sqrt{s_{NN}}$: the total baryon number (participants) is fixed, where it ends up in rapidity is not

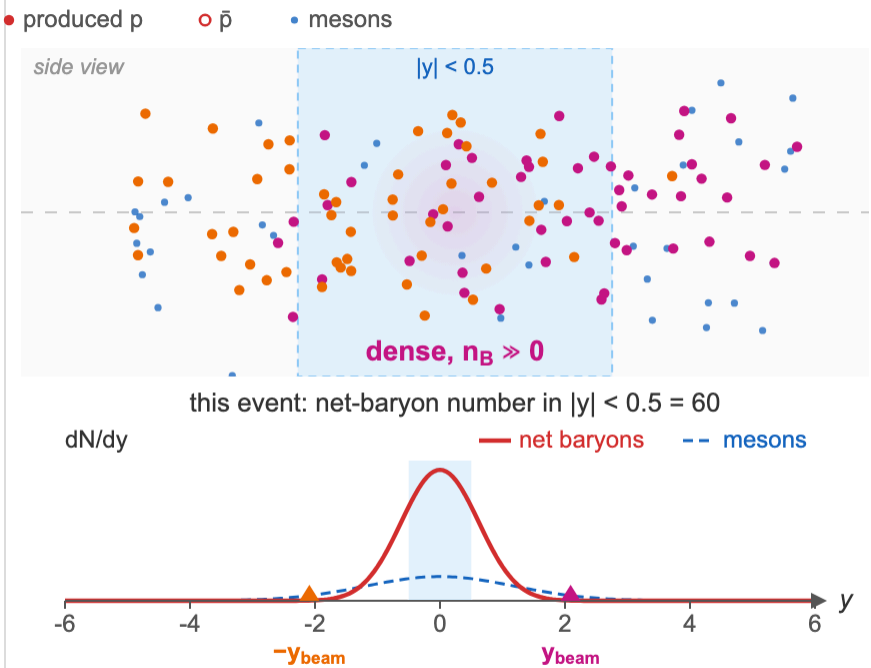
High energy $\sqrt{s_{NN}}$ = 200 GeV — e.g. RHIC top energy, LHC



Freeze-out: $\mu_B \approx 24$ MeV, $T \approx 166$ MeV

- Nuclei pass through each other (transparency)
- Mid-rapidity: matter made from the vacuum, $\bar{p}/p \approx 0.75$

Low energy $\sqrt{s_{NN}}$ = 7.7 GeV — e.g. RHIC BES-II, SPS



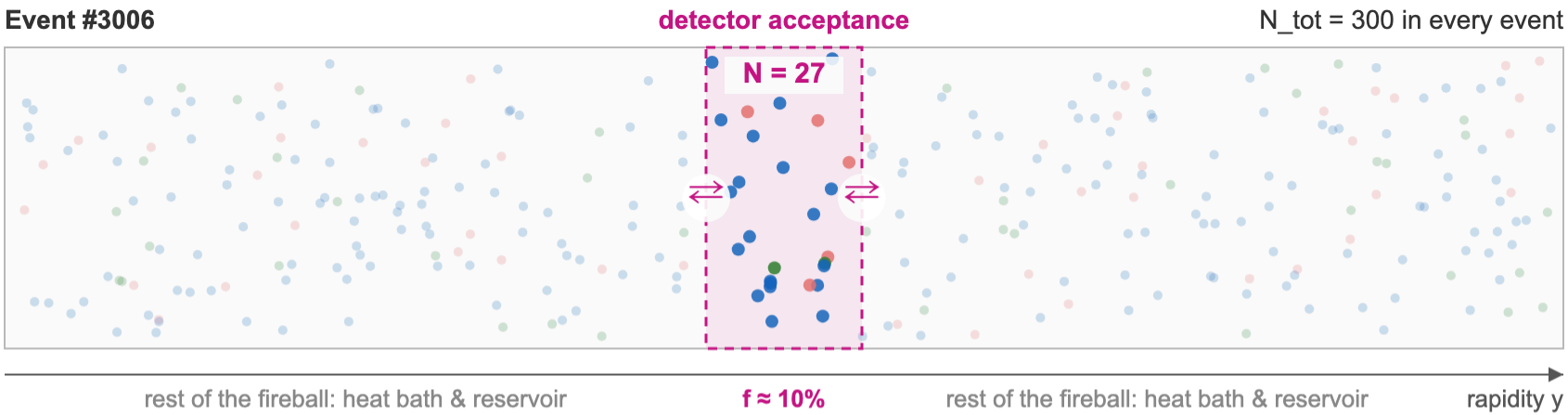
Freeze-out: $\mu_B \approx 420$ MeV, $T \approx 140$ MeV

- Nuclei are stopped: baryons pile up at mid-rapidity
- Mid-rapidity: compressed baryon-rich matter, $\bar{p}/p \ll 1$

Lower $\sqrt{s_{NN}}$ ⇒ more baryon stopping ⇒ larger n_B and μ_B at mid-rapidity ⇒ scanning $\sqrt{s_{NN}}$ scans the phase diagram along μ_B

Freeze-out $\mu_B(\sqrt{s_{NN}}), T(\mu_B)$: J. Cleymans, H. Oeschler, K. Redlich, S. Wheaton, hep-ph/0511094

Why grand canonical ensemble?

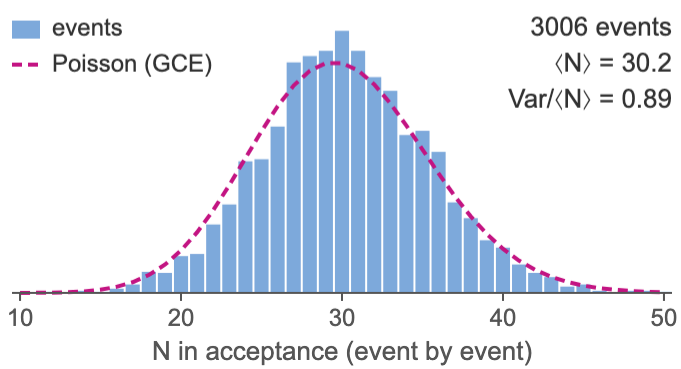


- Every event: total N and charges fixed (canonical)
- Detector sees only a fraction f ≪ 1 of all particles
- The rest is a heat bath & reservoir: exchanges E, Q

⇒ Grand canonical ensemble in the acceptance (T, μ)

- Var(N) / ⟨N⟩ = 1 − f → 1 (Poisson) for f ≪ 1

Requires Δy(corr), Δy(thermal) ≪ Δy(acc) ≪ Δy(total)

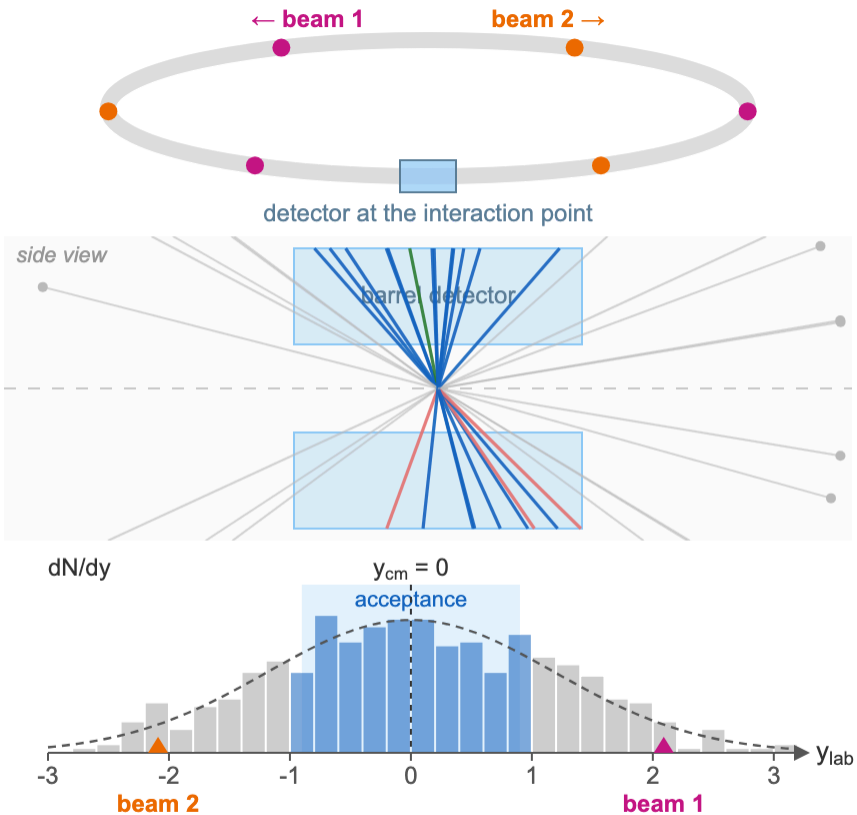


↻ Replay

Collider vs Fixed target

Same Au beam in both cases: E_{beam} = 3.85 GeV per nucleon, $\gamma \approx 4.1$, $y_{beam} \approx 2.09$

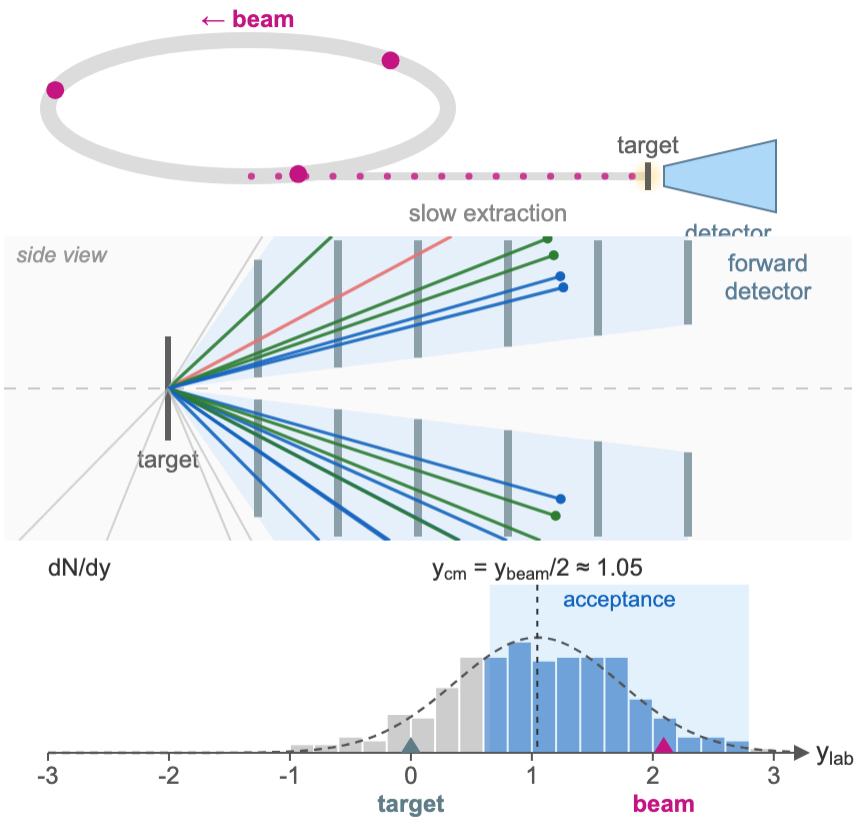
Collider e.g. RHIC (STAR BES), LHC



$\sqrt{s_{NN}} = 2E_{beam} = 7.7$ GeV

- CM = lab: symmetric around y = 0, mid-rapidity at 90°
- Low √s: luminosity drops (beams hard to focus)

Fixed target e.g. SIS18, SPS, STAR FXT, FAIR



$\sqrt{s_{NN}} = \sqrt{(2m_N^2 + 2m_N E_{beam})} = 3.0$ GeV

- CM boosted forward: acceptance in y_{cm} shifts with √s
- Dense target: high interaction rate

Scanning the phase diagram: experiments

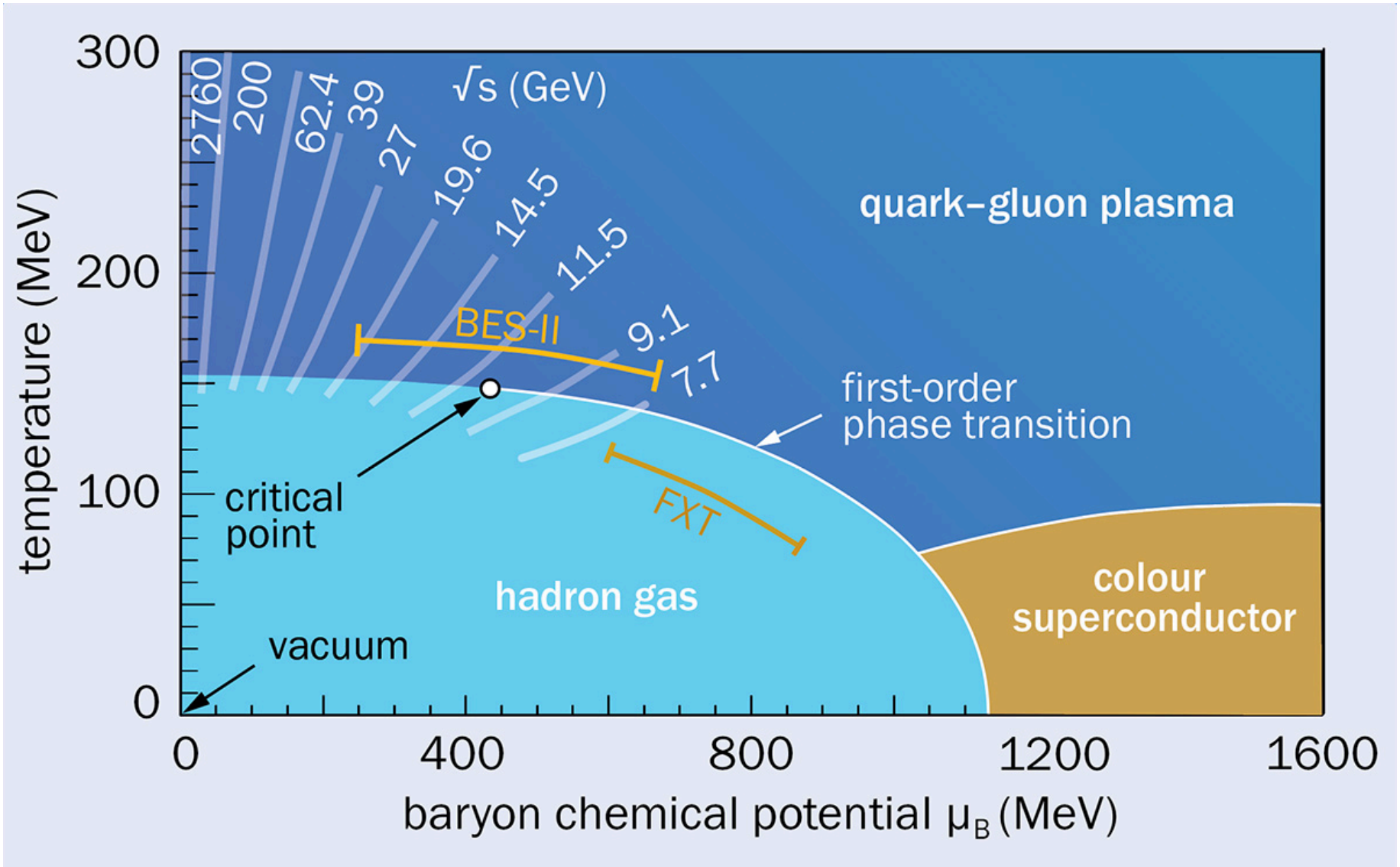
Experiment	Mode	$\sqrt{s_{NN}}$ [GeV]
STAR BES-II (RHIC)	collider	7.7 – 27
STAR FXT (RHIC)	fixed target	3.0 – 13.7
HADES (GSI)	fixed target	2.4 – 2.55
NA61/SHINE (CERN SPS)	fixed target	5.1 – 17.3
CBM (FAIR, future)	fixed target	2.9 – 4.9
MPD (NICA, future)	collider	4 – 11

Lower $\sqrt{s_{NN}} \Rightarrow$ larger μ_B at freeze-out: the beam energy scan scans the phase diagram along the freeze-out curve

Outline

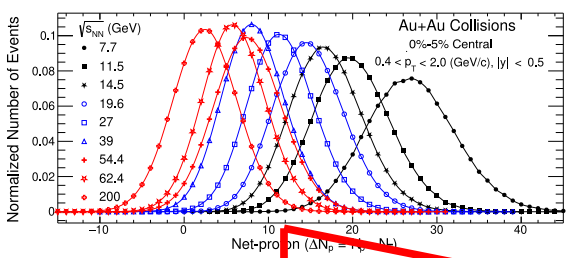
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QCD phase diagram



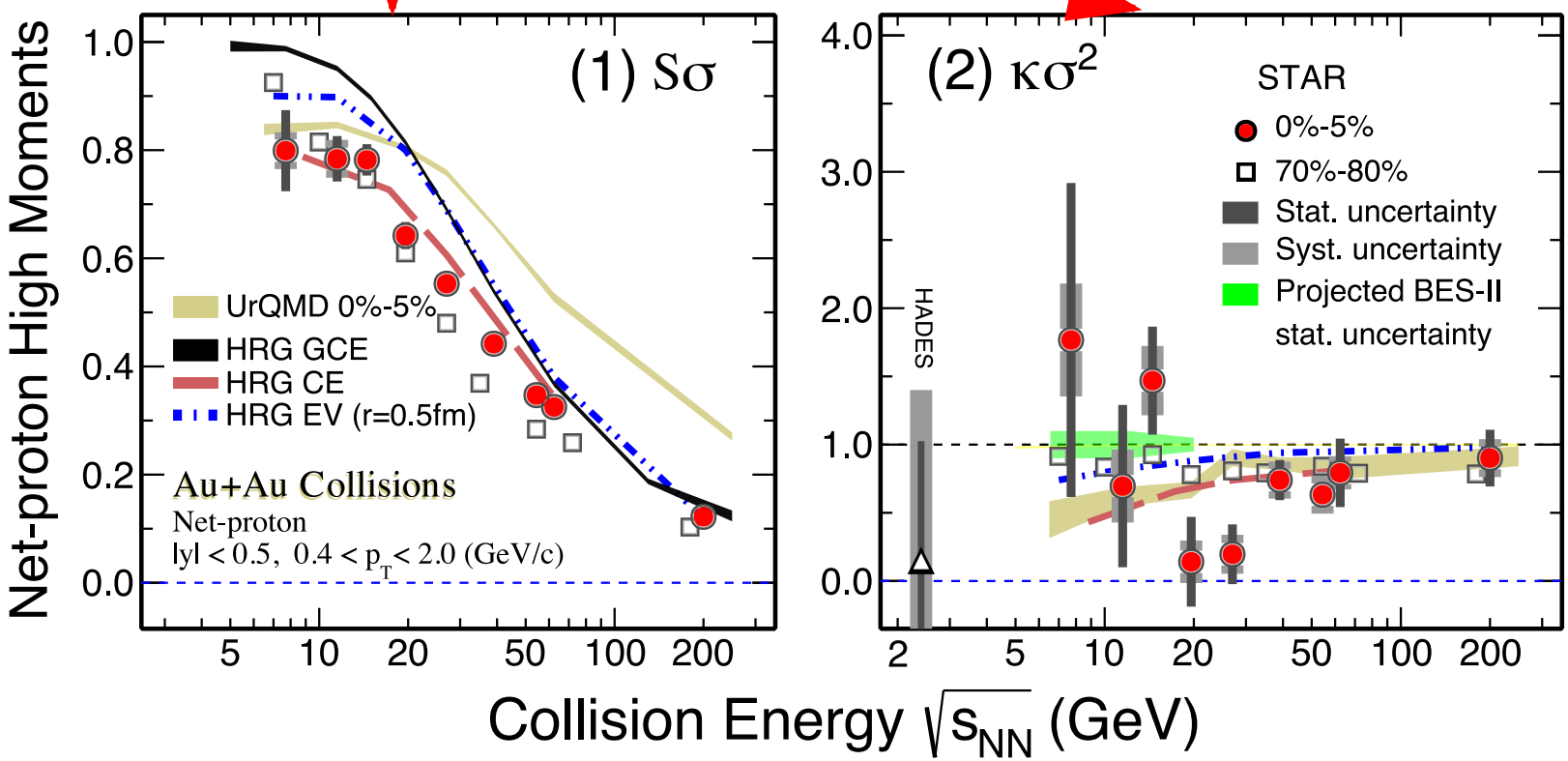
Lipei Du, Agnieszka Sorensen, Mikhail Stephanov, arXiv:2402.10183

Cumulants of the net-proton distribution



$$\delta N_B = N_B - \langle N_B \rangle$$
$$C_1 = \langle N_B \rangle$$
$$C_2 = \langle (\delta N_B)^2 \rangle$$
$$C_3 = \langle (\delta N_B)^3 \rangle$$
$$C_4 = \langle (\delta N_B)^4 \rangle - 3\langle (\delta N_B)^2 \rangle^2$$

$$S\sigma = C_3/C_2$$
$$\kappa\sigma^2 = C_4/C_2$$



STAR Collaboration, arXiv:2001.02852

Why higher-order cumulants?

$$\chi_n^B = \frac{\partial^n \left(p/T^4\right)}{\partial(\mu_B/T)^n}, \qquad C_n = VT^3\chi_n^B, \qquad R_{nm}^B = \frac{\chi_n^B}{\chi_m^B} : \quad S\sigma = R_{32}^B, \quad \kappa\sigma^2 = R_{42}^B$$

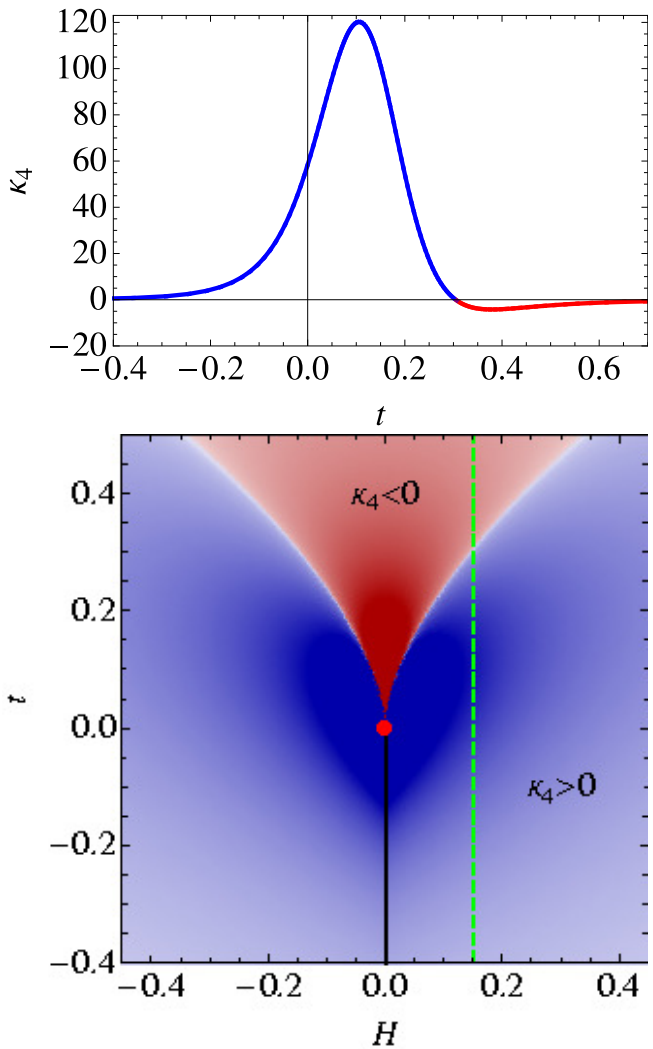
- Cumulants = derivatives of the QCD pressure: direct link between theory and measurements
- Ratios cancel the unknown volume V; baseline without correlations (Skellam / HRG): $R_{42}^B = 1$
- Near the CEP higher orders grow with higher powers of the correlation length:

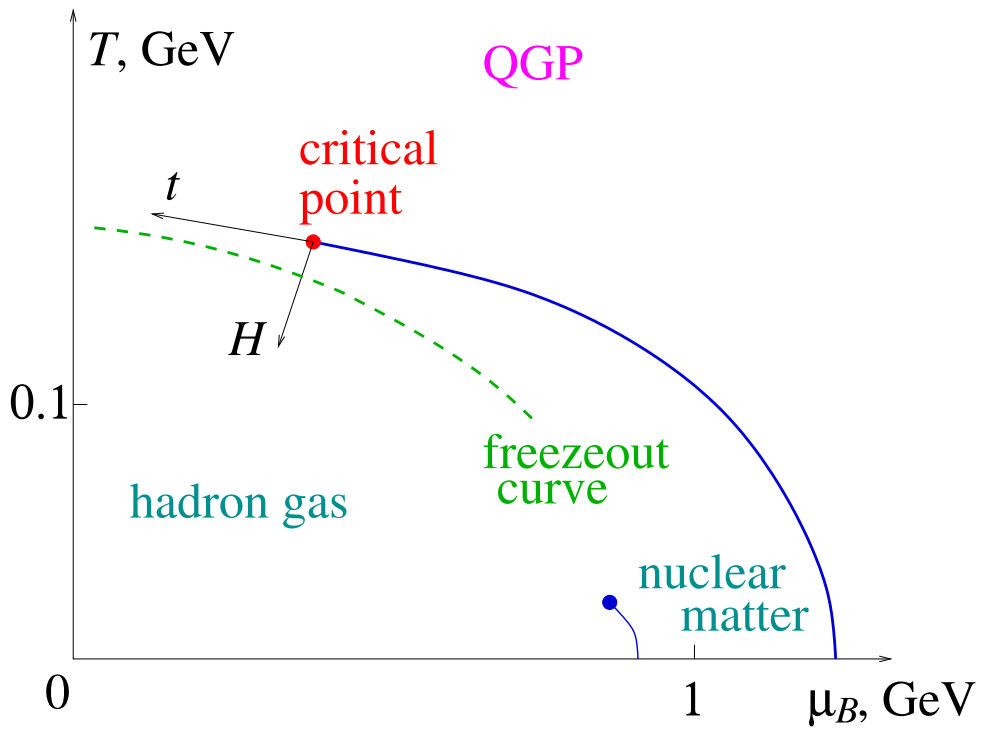
$$C_2 \sim \xi^2, \qquad C_3 \sim \xi^{4.5}, \qquad C_4 \sim \xi^7$$

- Caveats: net-proton vs net-baryon, finite size and time ($\xi \lesssim 2$–3 fm), baryon number conservation

M. A. Stephanov, Phys. Rev. Lett. 102, 032301 (2009), arXiv:0809.3450

Non-monotonic behavior of 4th order cumulants





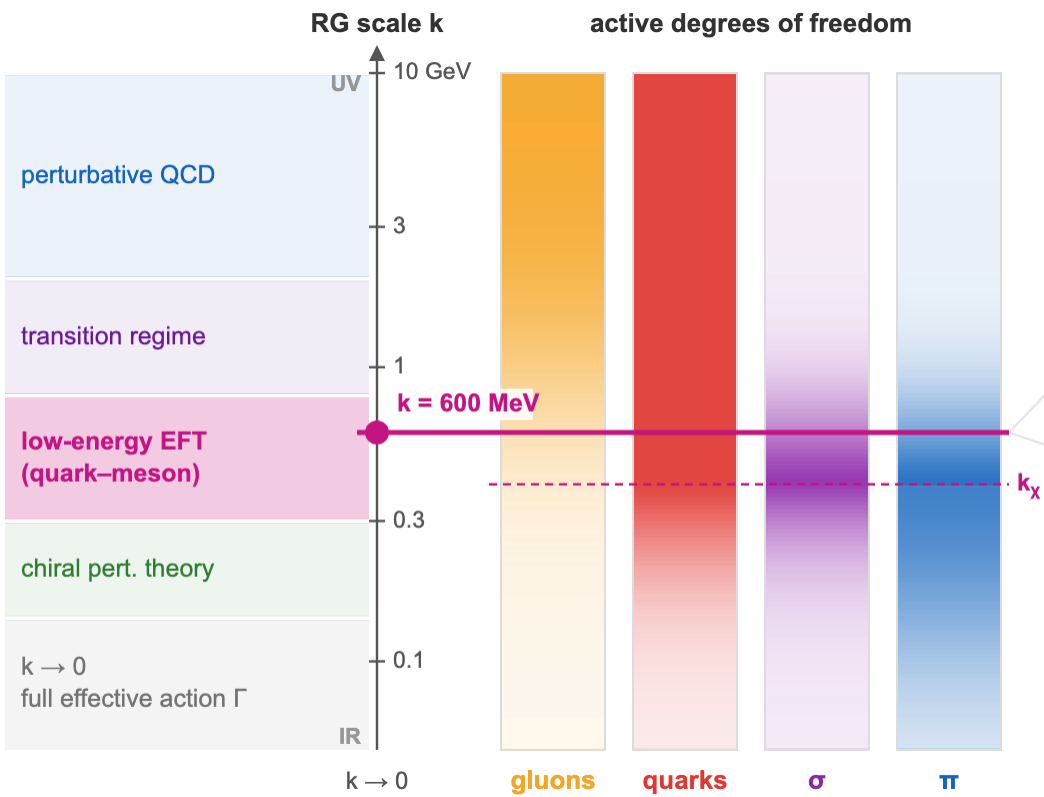
M. A. Stephanov, arXiv:1104.1627

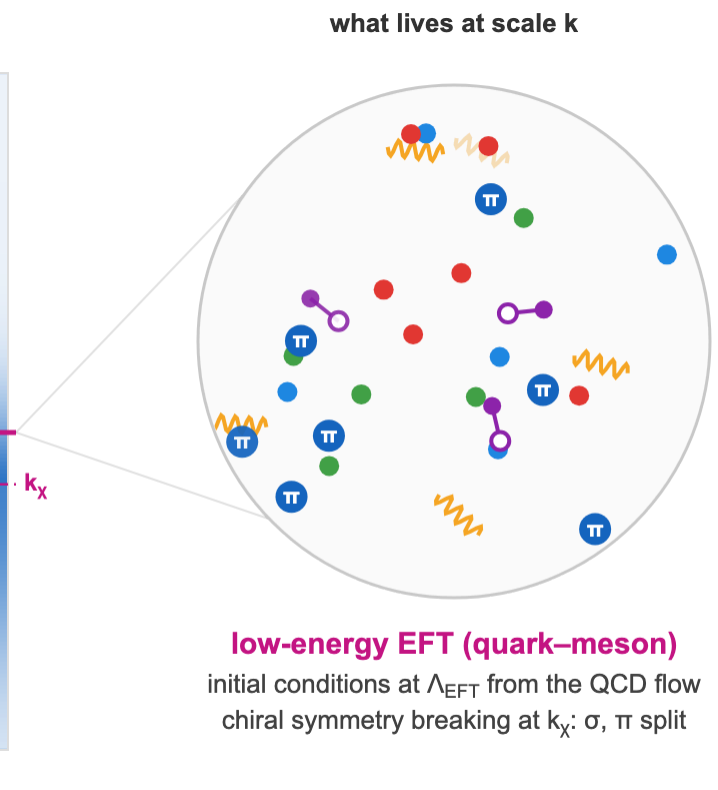
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From QCD to low-energy effective theories

- Assume the system is in a state of equilibrium
- Assume the system can be described by the grand canonical ensemble





Low energy effective field theory (LEFT)

- Linear sigma model (only mesons)

  Suitable for describing the scaling behavior and universality class

- NJL model (only quarks)

  Suitable for describing the chiral symmetry breaking and the color superconductivity

- (Polyakov-) Quark-Meson model (mesons+quarks)

  The ideal framework for phenomenology, bridging critical fluctuations and thermal dynamics

Low energy effective field theory (LEFT)

- Polyakov-Quark-Meson (PQM) model within functional renormalization group (fRG)

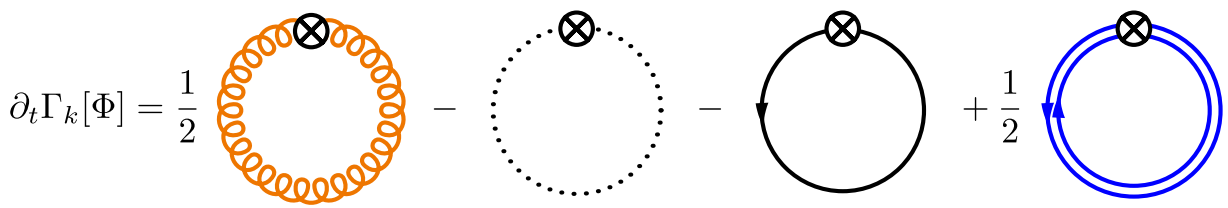
$$\Gamma_k[\phi, q, \bar{q}] = \int_x \Big\{ Z_{q,k} \bar{q} \Big[\gamma_\mu \partial_\mu - \gamma_0 (\mu + i g A_0) \Big] q + \frac{1}{2} Z_{\phi,k} (\partial_\mu \phi)^2 \\ + h_k \, \bar{q} \, \left(\tau^0 \sigma + \boldsymbol{\tau} \cdot \boldsymbol{\pi} \right) \, q + V_k(\rho, A_0) - c\sigma \Big\}$$

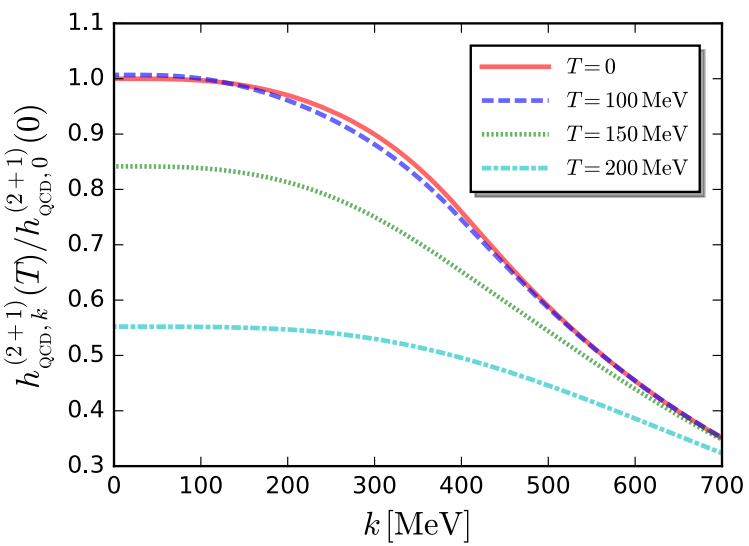
- Baryon number fluctuations

$$\Omega[T, \mu_B] = V_{k=0}(\rho, A_0) - c\sigma \Big|_{\mathrm{EoM}}, \qquad p = -\Omega[T, \mu_B]$$

$$\chi_n^B = \frac{\partial^n}{\partial (\mu_B/T)^n} \, \frac{p}{T^4} \, , \qquad R_{nm}^B = \frac{\chi_n^B}{\chi_m^B}$$

QCD-assisted LEFT: input from fRG-QCD





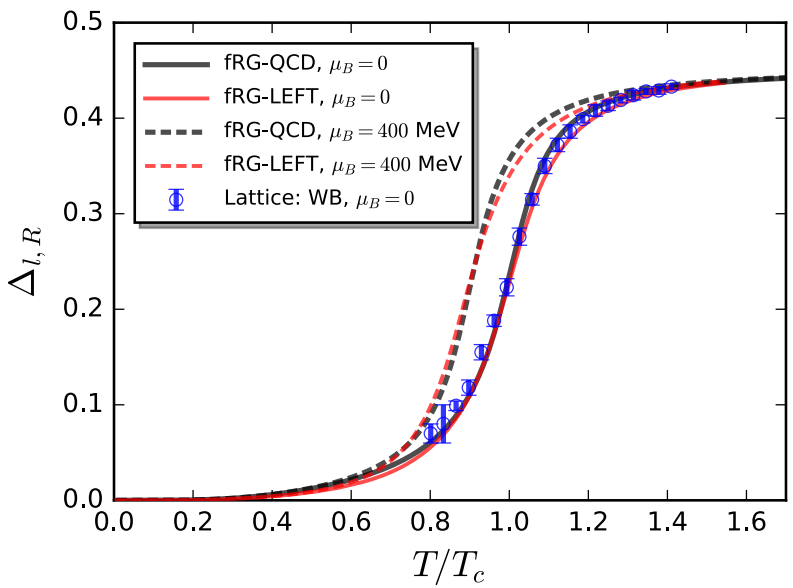
- LEFT = QCD flow below $\Lambda = 700$ MeV: the glue sector decouples (gluon mass gap)
- Thermal Yukawa coupling from 2+1 flavor fRG-QCD:

$$h_k(T) = h_0 \, \frac{h_k^{\mathrm{QCD}}(T)}{h_0^{\mathrm{QCD}}(0)}$$

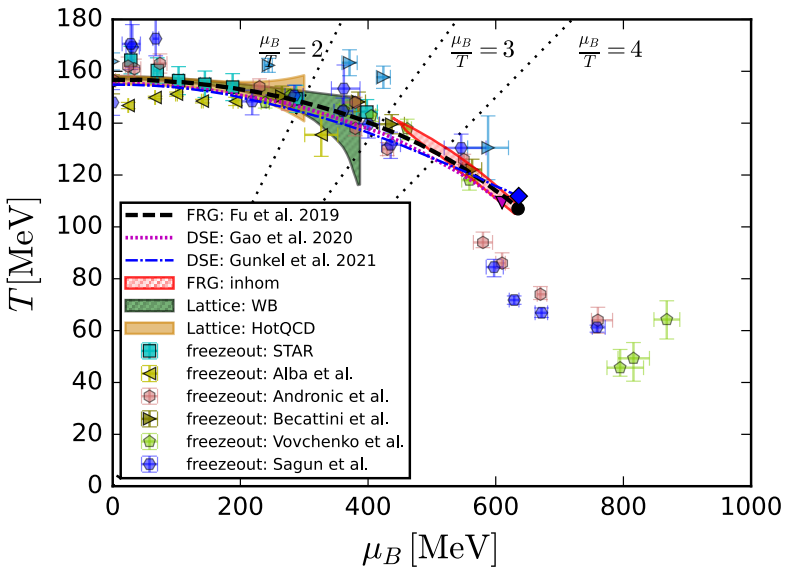
- Glue background: Polyakov loop potential from Yang–Mills, rescaled to QCD
- Remaining parameters $\nu_\Lambda,\ \lambda_\Lambda,\ h_0,\ c_\sigma$ fixed by vacuum observables

Fu, Luo, Pawlowski, Rennecke, **SY**, arXiv:2308.15508; Fu, Pawlowski, Rennecke, arXiv:1909.02991

Benchmark and phase structure



Chiral condensate: agrees with lattice and fRG-QCD

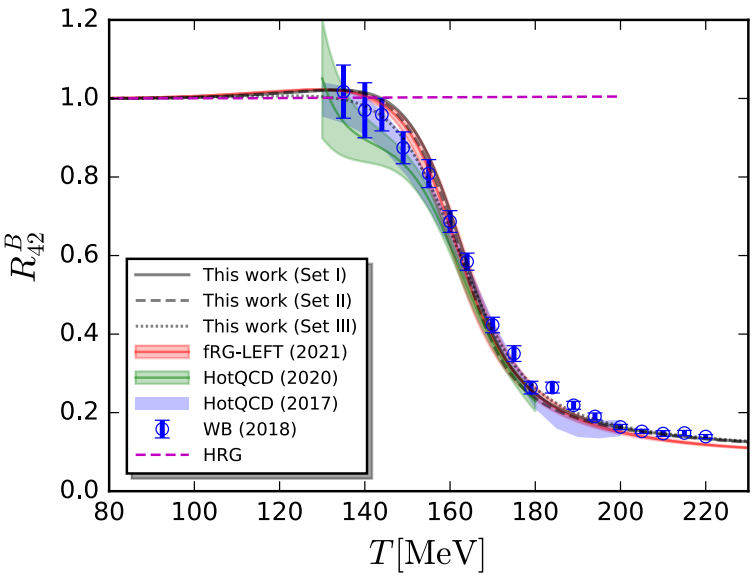


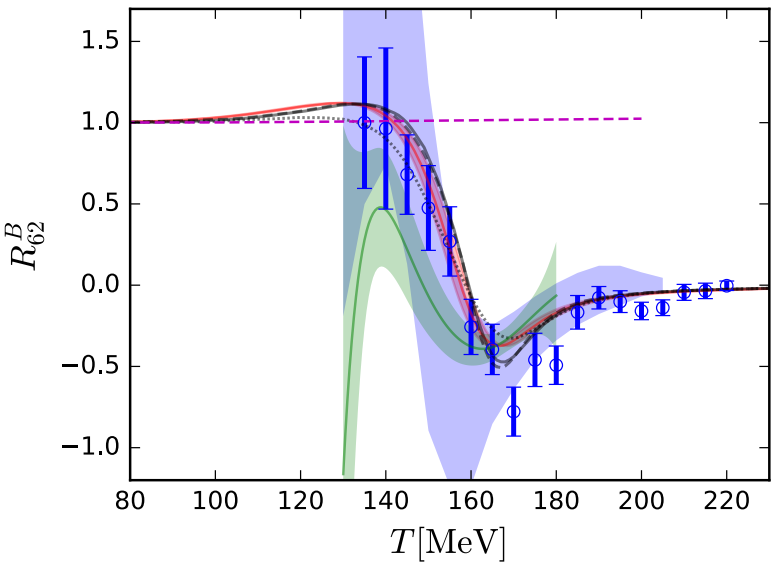
CEP at $(T, \mu_B)_{\mathrm{CEP}} = (98,\ 643)$ MeV

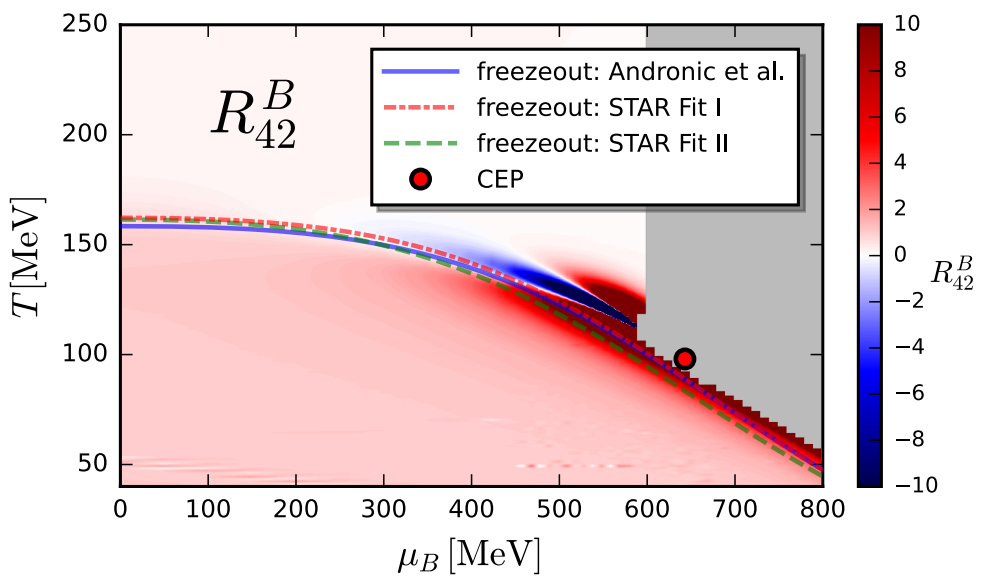
QCD input moves the CEP into the functional QCD range (600–650 MeV); conventional (P)QM models: $\mu_{B,\mathrm{CEP}} > 850$ MeV

Fu, Luo, Pawlowski, Rennecke, **SY**, arXiv:2308.15508

Predictions of the fluctuations

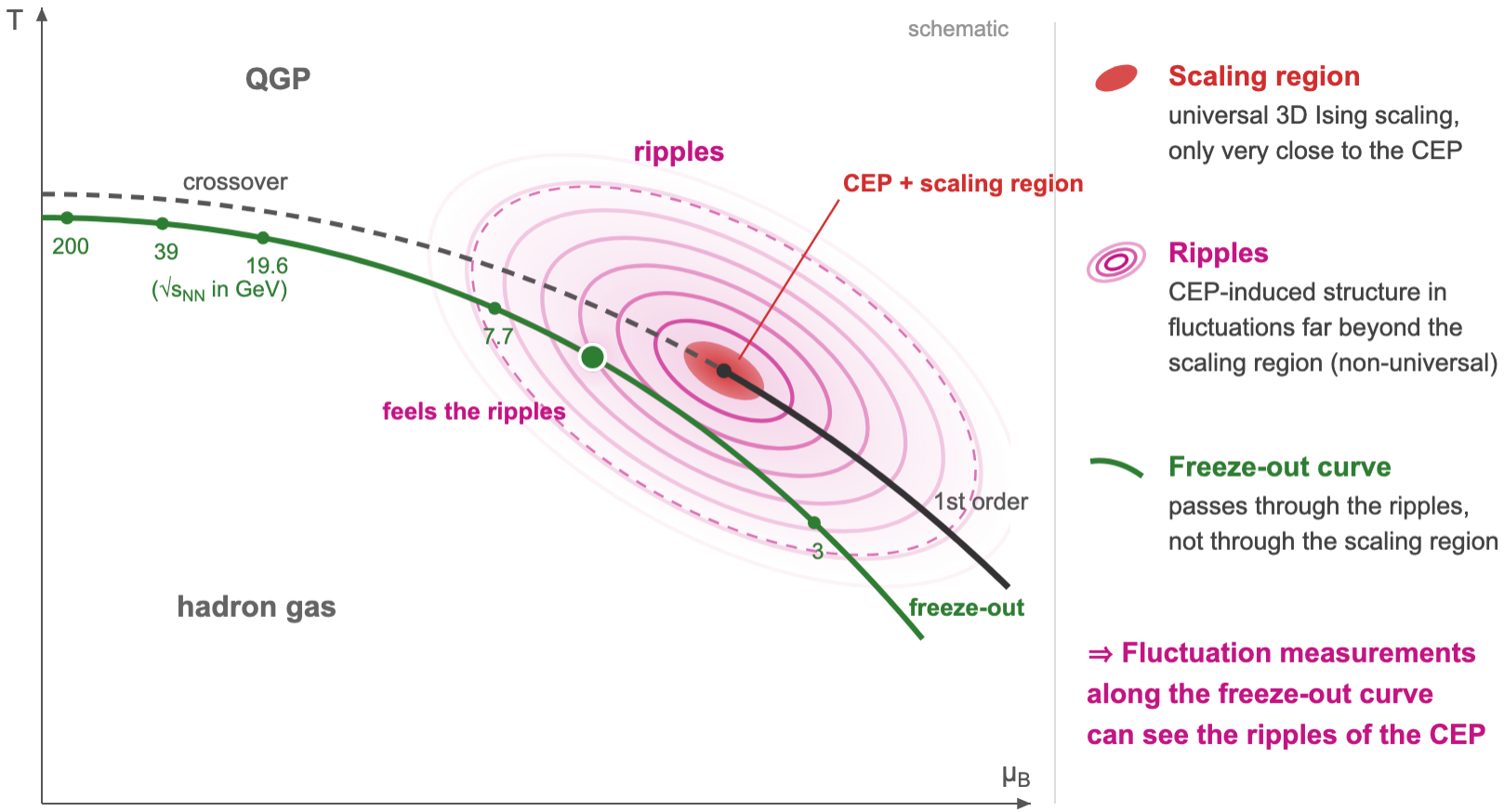






W.-j. Fu, X. Luo, J. M. Pawlowski, F. Rennecke, **SY**, arXiv:2308.15508

Ripples of the QCD critical point



Scaling region
universal 3D Ising scaling, only very close to the CEP

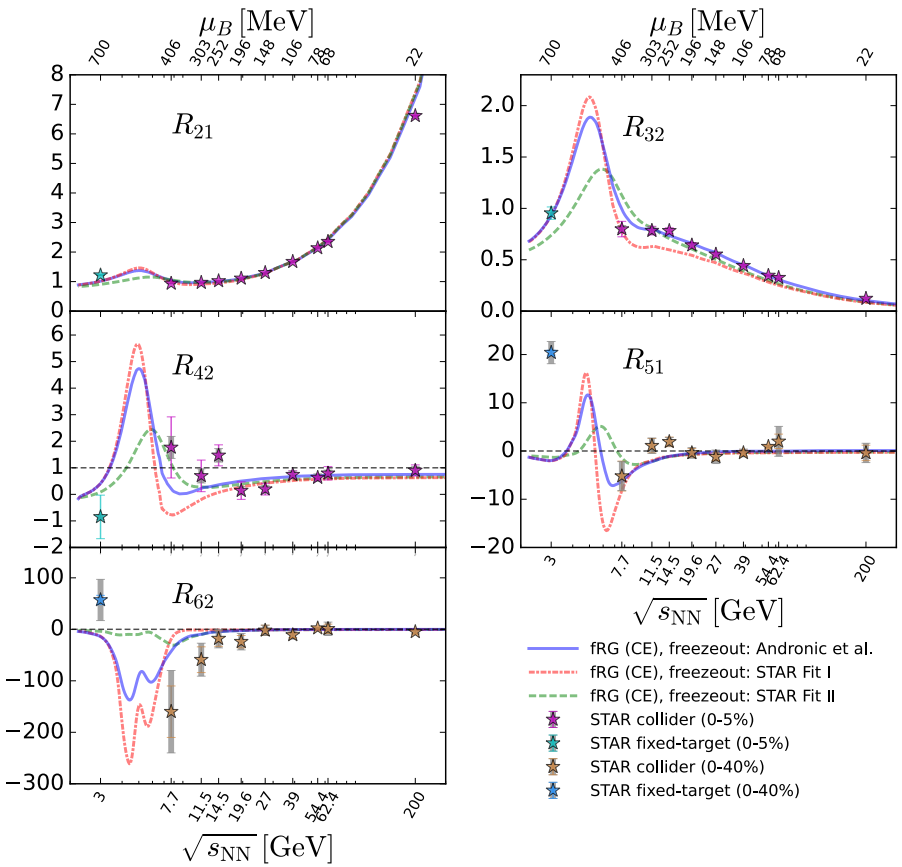
Ripples
CEP-induced structure in fluctuations far beyond the scaling region (non-universal)

Freeze-out curve
passes through the ripples, not through the scaling region

⇒ Fluctuation measurements along the freeze-out curve can see the ripples of the CEP

W.-j. Fu, X. Luo, J. M. Pawlowski, F. Rennecke, **SY**, arXiv:2308.15508

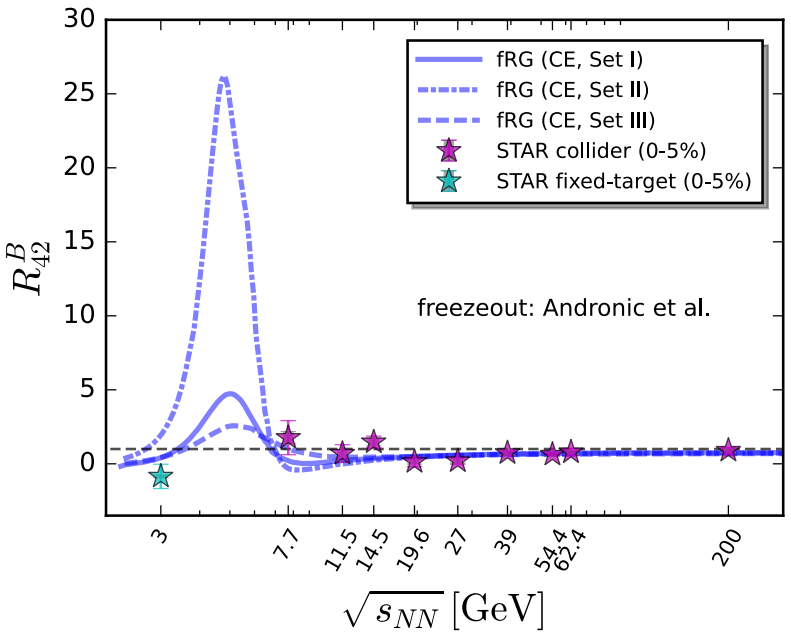
Fluctuations along the freeze-out curve

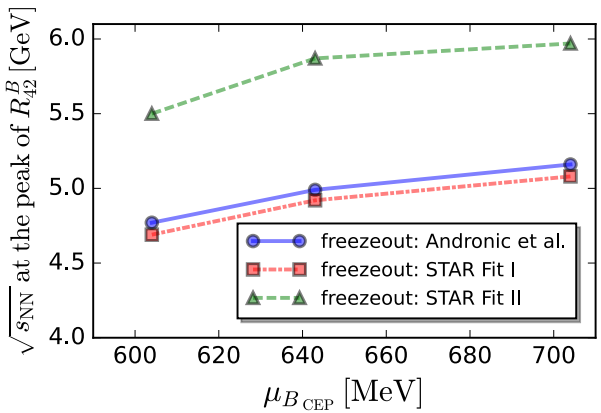


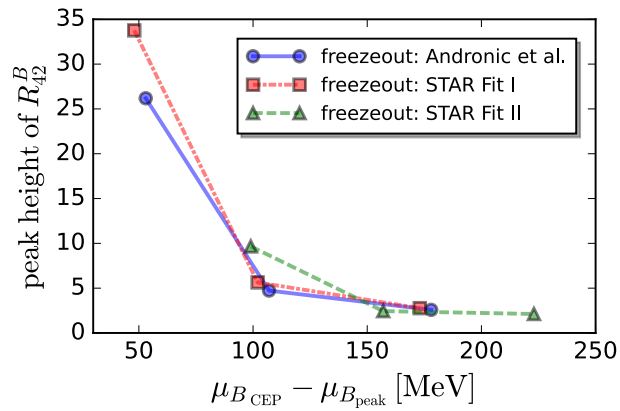
- Global baryon number conservation (canonical corrections) included for $\sqrt{s_{NN}} \lesssim 11.5$ GeV
- Non-monotonic R^B_{42} for 7.7–200 GeV, consistent with STAR
- Prominent **peak** at $3 \lesssim \sqrt{s_{NN}} \lesssim 7.7$ GeV: the fixed-target range
- Hyper-order R^B_{51}, R^B_{62} turn negative at lower $\sqrt{s_{NN}}$

Fu, Luo, Pawlowski, Rennecke, **SY**, arXiv:2308.15508

What does the peak tell us?







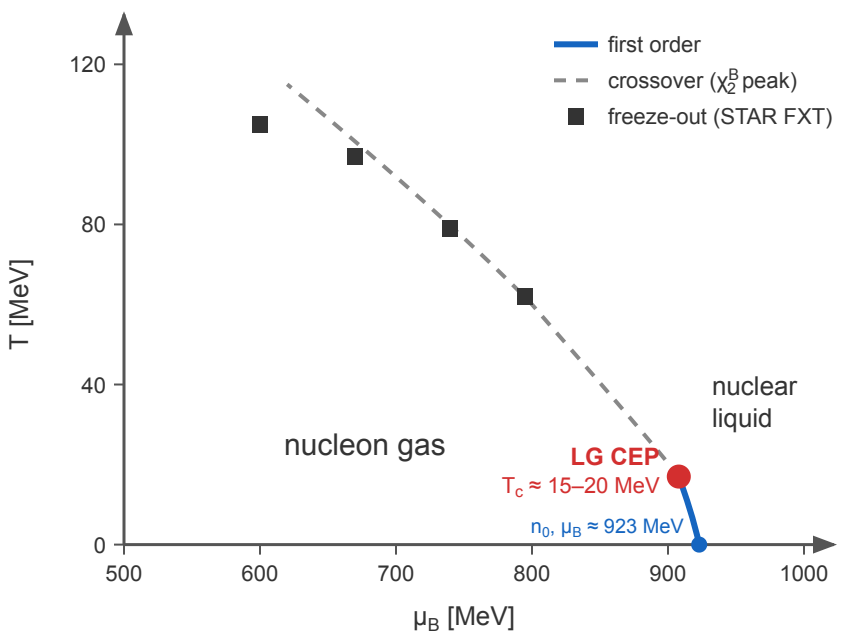
- Peak **position** ($\sqrt{s_{NN}} \approx 5$–6 GeV) is set mainly by the freeze-out curve
- Peak **height** depends on $\mu_{B,\mathrm{CEP}} - \mu_{B,\mathrm{peak}}$, growing strongly below ∼100 MeV

The height of the peak encodes the location of the CEP

Outline

- QCD phase diagram, phase transitions and universality
- Heavy-ion collisions
- Baryon number fluctuations
- QCD-assisted low energy effective theory
- Fluctuations near the liquid–gas transition
- Upgraded fRG computations and critical behavior
- Summary and Outlook

The nuclear liquid–gas transition



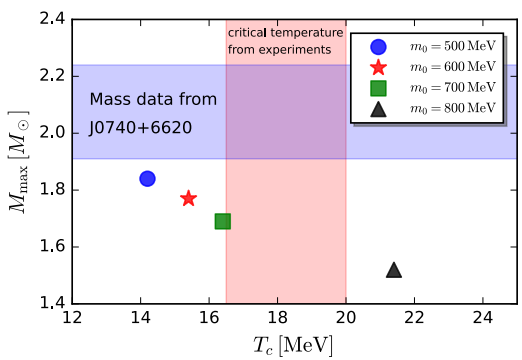
- $T = 0$: nuclear matter appears at $\mu_B \approx m_N - 16$ MeV ≈ 923 MeV (saturation density $n_0 \approx 0.16\ \mathrm{fm}^{-3}$)
- First-order line between nucleon gas and nuclear liquid, ending at a CEP with $T_c \approx 15$–20 MeV
- Same Z_2 (3D Ising) universality class as the QCD CEP
- Low-energy collisions ($\sqrt{s_{NN}} \lesssim 4$ GeV) freeze out close to this region

Parity-doublet model (PDM)

$$\mathcal{L}_F = \bar{N}_1\big(\gamma_\mu\partial_\mu + g_1(\sigma + i\gamma_5\vec{\tau}\cdot\vec{\pi})\big)N_1 + \bar{N}_2\big(\gamma_\mu\partial_\mu + g_2(\sigma - i\gamma_5\vec{\tau}\cdot\vec{\pi})\big)N_2$$
$$+ m_0\big(\bar{N}_1\gamma_5 N_2 - \bar{N}_2\gamma_5 N_1\big)$$

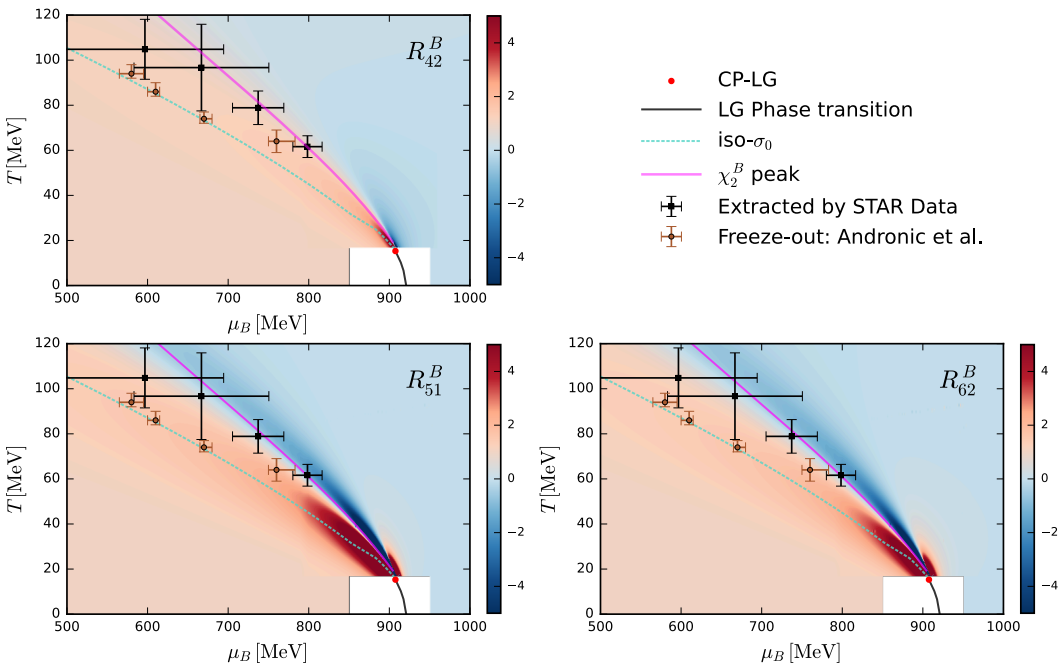
$$m_\pm = \tfrac{1}{2}\sqrt{(g_1+g_2)^2\sigma^2 + 4m_0^2} \pm \tfrac{g_1-g_2}{2}\,\sigma$$

- Nucleon $N(939)$ and its parity partner $N^*(1535)$; chirally invariant mass m_0
- ω exchange: repulsion; parameters fitted to nuclear matter at saturation
- $m_0 = 600$ MeV from LG critical temperature and neutron star maximum mass
- Mean field, grand canonical; reliable for $T \lesssim 120$ MeV



Recchi and **SY**, arXiv:2609.17328

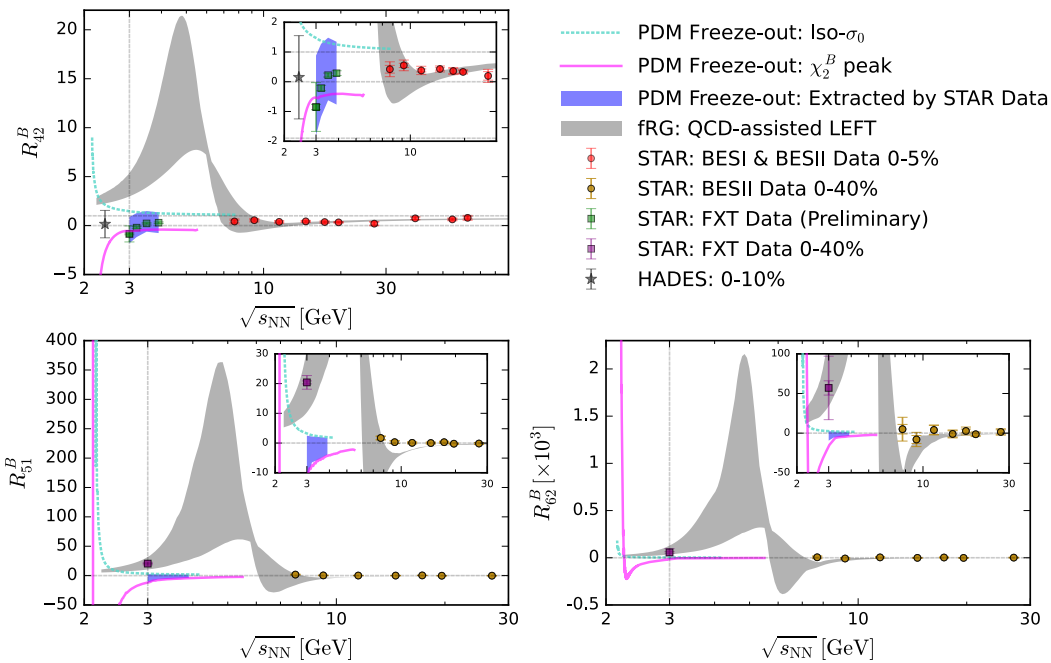
Fluctuations near the liquid–gas transition



- Dilute gas: $R^B_{42} = 1$ (Skellam)
- Strong sign changes of R^B_{42}, R^B_{51}, R^B_{62} toward the LG critical point
- Freeze-out at high μ_B is poorly known. Three scenarios:
  - constant $\sigma_0 = 88$ MeV
  - peak of χ_2^B
  - extracted from R_{21}, R_{31} (STAR FXT)

Recchi and **SY**, arXiv:2609.17328

Comparison with low-energy data



- FXT data below 7.7 GeV: $R_{42} < 1$
- QCD-assisted LEFT (quarks + mesons, gray): $R_{42}^B \to 1$ at low energy
- PDM with freeze-out extracted from R_{21}, R_{31} (blue) agrees with preliminary STAR FXT data for $\sqrt{s_{NN}} \lesssim 4$ GeV
- Not yet included: non-equilibrium, baryon number conservation, protons vs neutrons

Nucleon interactions and the LG transition may contribute significantly to fluctuations at low collision energies

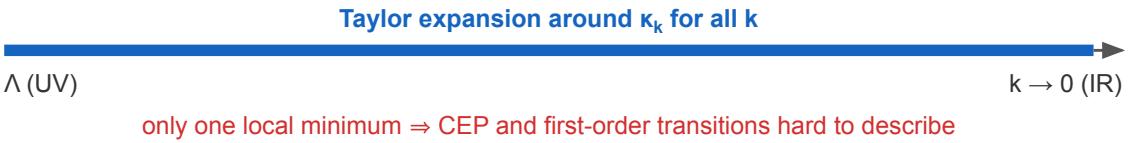
Recchi and **SY**, arXiv:2609.17328

Outline

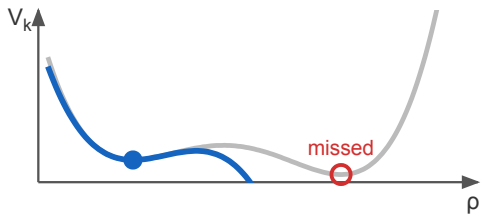
- QCD phase diagram, phase transitions and universality
- Heavy-ion collisions
- Baryon number fluctuations
- QCD-assisted low energy effective theory
- Fluctuations near the liquid–gas transition
- Upgraded fRG computations and critical behavior
- Summary and Outlook

Upgrading the fRG computation

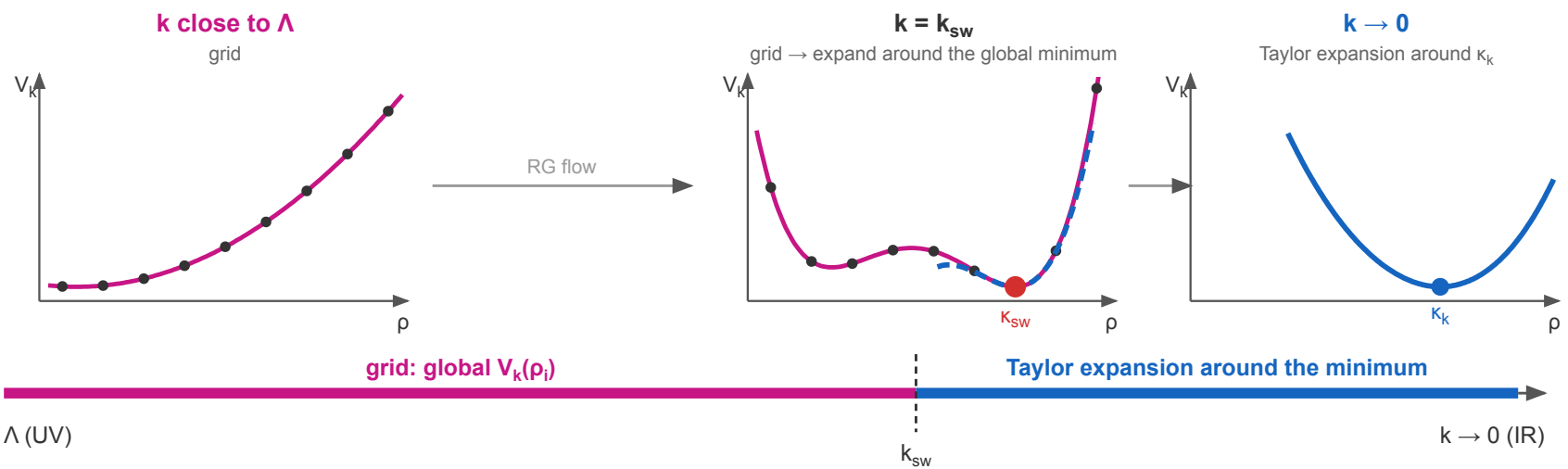
Before: Taylor expansion only



only one local minimum ⇒ CEP and first-order transitions hard to describe



Now: grid → Taylor expansion



$$k > k_{\mathrm{sw}}: \quad V_k(\rho_i), \quad i = 1, \ldots, N \qquad k \leq k_{\mathrm{sw}}: \quad V_k(\rho) = \sum_{n=0}^{N} \frac{\lambda_{n,k}}{n!} (\rho - \kappa_k)^n, \quad \lambda_{n,k_{\mathrm{sw}}} = V_{k_{\mathrm{sw}}}^{(n)}(\kappa_{\mathrm{sw}})$$

- Grid ($\Lambda \to k_{\mathrm{sw}}$): global shape of V_k, true minimum, first-order transitions and the CEP
- Taylor ($k_{\mathrm{sw}} \to 0$): smooth flow into the deep IR, accurate derivatives for high-order fluctuations

SY, in preparation

Computing high-order fluctuations

Chain rule at the saddle point

$$p = -\Omega\big(\mu_B; \Phi_0(\mu_B)\big), \qquad \left.\frac{\partial \Omega}{\partial \Phi}\right|_{\Phi_0} = 0$$

$$\Phi = (\sigma, L, \bar{L}), \qquad \frac{d\Omega}{d\mu_B} = \Omega_\mu$$

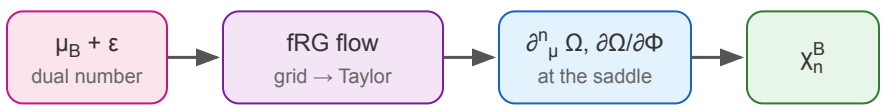
$$\frac{d^2\Omega}{d\mu_B^2} = \Omega_{\mu\mu} + \Omega_{\mu\Phi}\,\frac{d\Phi_0}{d\mu_B}, \qquad \frac{d\Phi_0}{d\mu_B} = -\,\Omega_{\Phi\Phi}^{-1}\,\Omega_{\Phi\mu}$$

- Higher orders follow recursively from the chain rule
- Only partial derivatives of Ω at the saddle point are needed: **no finite differences** in μ_B

Algorithmic differentiation of the flow

$$\partial_t g_k = \beta_g[g_k; \mu_B] \quad \Longrightarrow \quad \partial_t\, \partial_{\mu_B}^n g_k = \partial_{\mu_B}^n \beta_g[g_k; \mu_B]$$

$$g_k \in \{\eta_\phi,\, \eta_q,\, \dots\} \quad \text{(all flows except } V_k\text{)}$$



- All μ_B-derivatives are propagated through the flow solver together with the couplings
- Exact to machine precision, no step size to tune

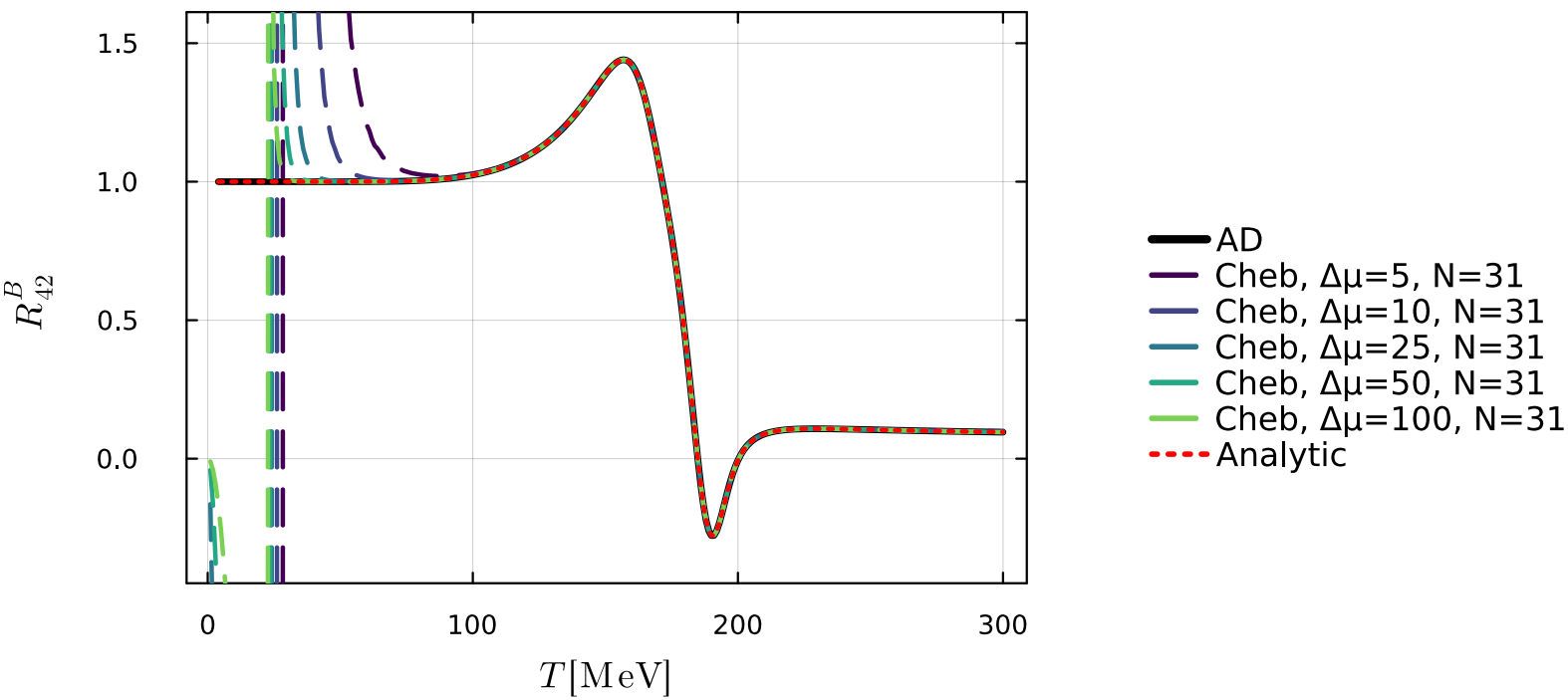
Chain rule: effective potential only | **Algorithmic differentiation**: all other flow equations (e.g. anomalous dimensions)

Stable high-order cumulants χ_n^B, also close to the CEP and the first-order line

SY, in preparation

Accuracy of the fluctuations

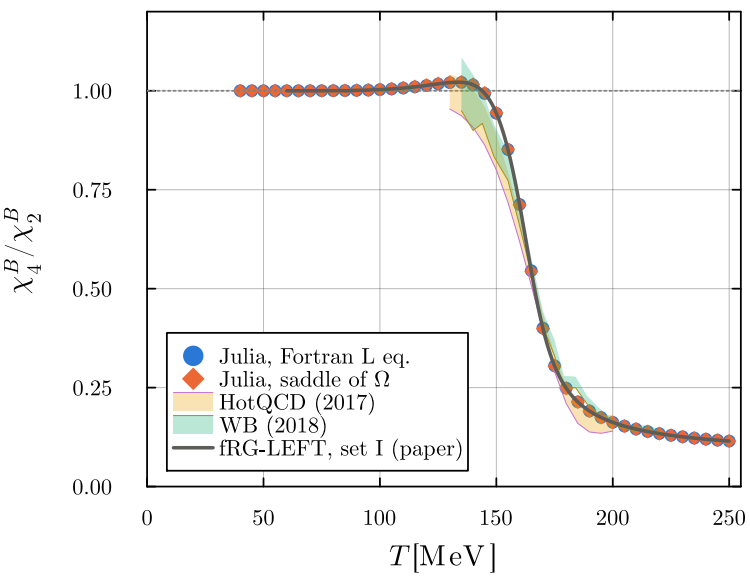
R^B_{42} at $\mu_B = 100$ MeV: AD vs Chebyshev vs analytic

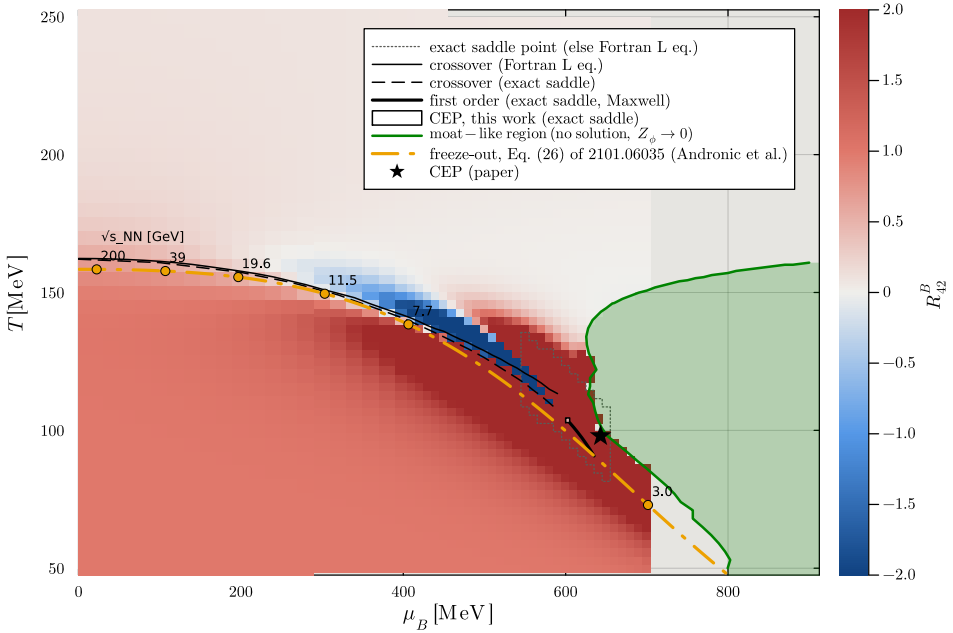


- AD coincides with the analytic result at all temperatures
- Chebyshev fits in μ_B ($\Delta\mu = 5$–100 MeV, $N = 31$) break down at low T, where $\chi^B_2 \to 0$

SY, in preparation

Improved baryon number fluctuations

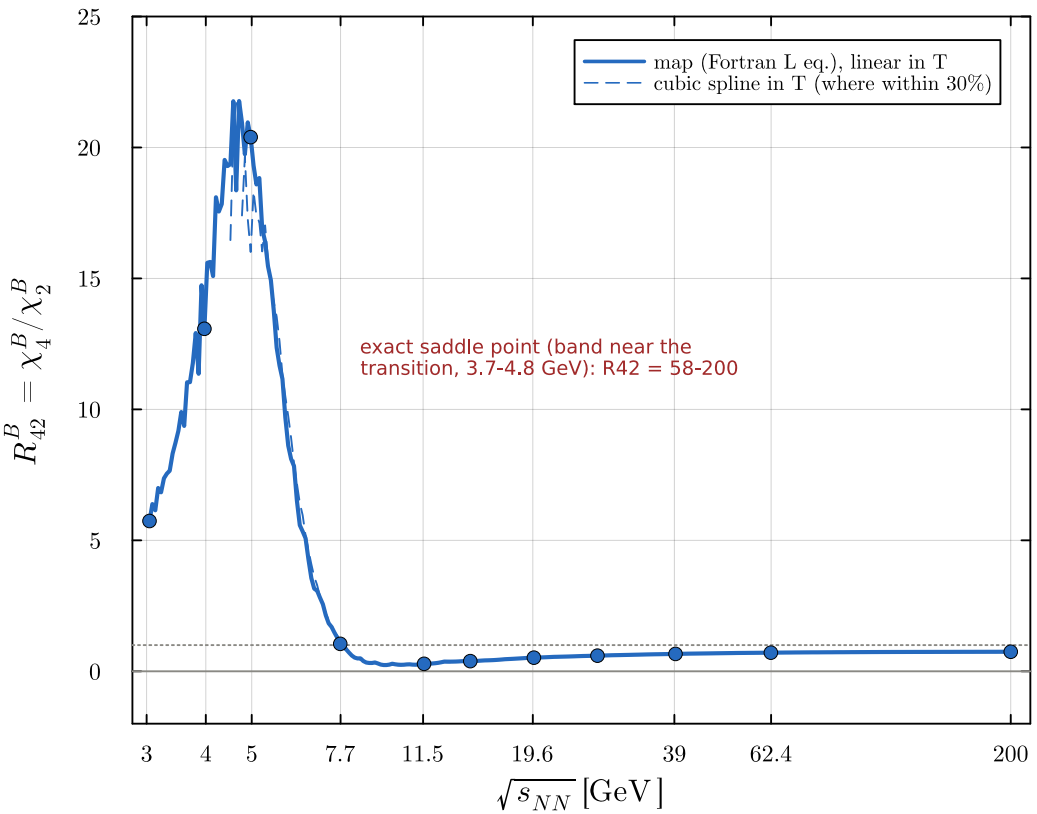




- $\mu_B = 0$: consistent with the previous computation
- Main improvement in the vicinity of the CEP
- Larger curvature near the first-order line: presumably due to missing high-density degrees of freedom (vector mesons, ω) and a Polyakov loop not computed self-consistently

SY, in preparation

Fluctuations on the freeze-out curve

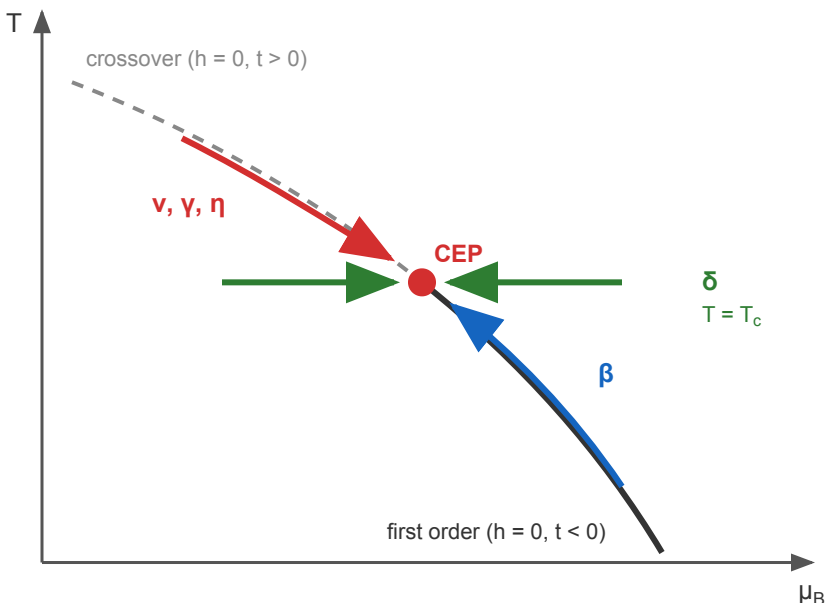


- R_{42}^B along the freeze-out curve, improved QCD-assisted LEFT (set I)
- Pronounced peak at $\sqrt{s_{NN}} \approx 4$–5 GeV: the fixed-target range
- Band near the transition (3.7–4.8 GeV): exact saddle point gives $R_{42}^B \approx 58$–200
- $\sqrt{s_{NN}} \geq 7.7$ GeV: $R_{42}^B < 1$, smooth rise towards 200 GeV

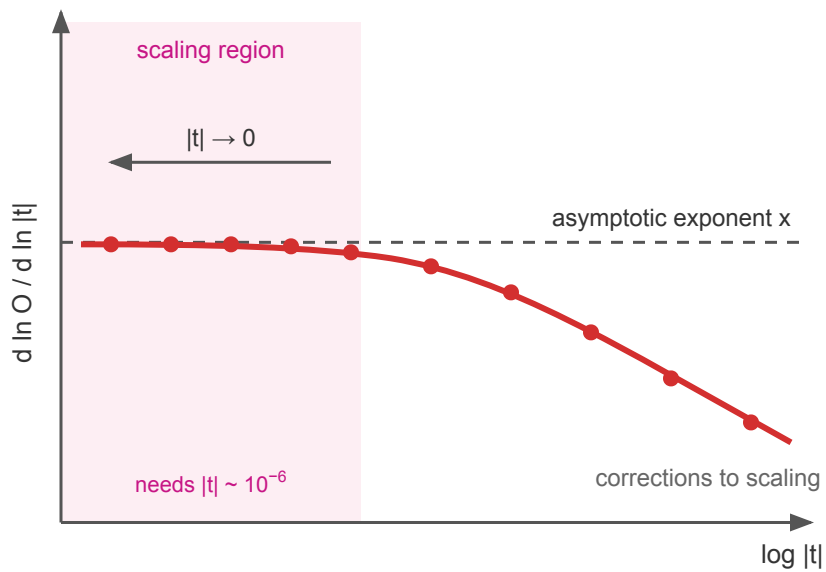
SY, in preparation

Extracting critical exponents at the CEP

Three directions towards the CEP



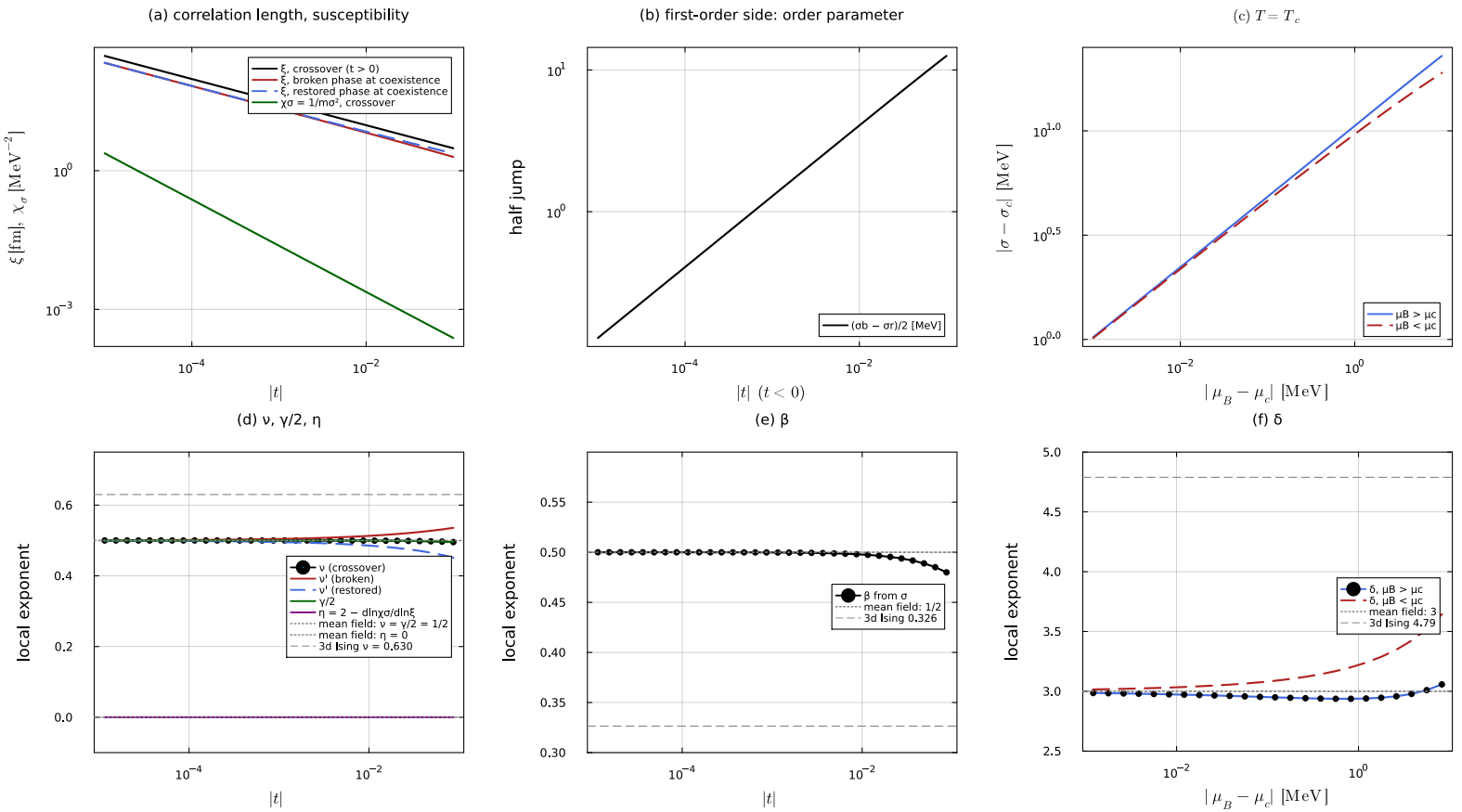
Local exponent → plateau



$$m_\sigma = \xi^{-1} \propto t^{\nu}, \quad \chi \propto t^{-\gamma}, \quad Z_{\mathrm{IR}} \propto t^{-\nu\eta} \qquad \tfrac{1}{2}\Delta\sigma \propto (-t)^{\beta} \qquad |\sigma - \sigma_c| \propto |\mu_B - \mu_c|^{1/\delta}$$

- Locate the CEP (T_c, μ_c) precisely, then approach it along the crossover, the first-order line and the critical isotherm
- Local exponents $d\ln O / d\ln|t|$ reach a plateau only very close to the CEP: requires the high-precision framework

QM model: Mean-field critical exponents at the CEP

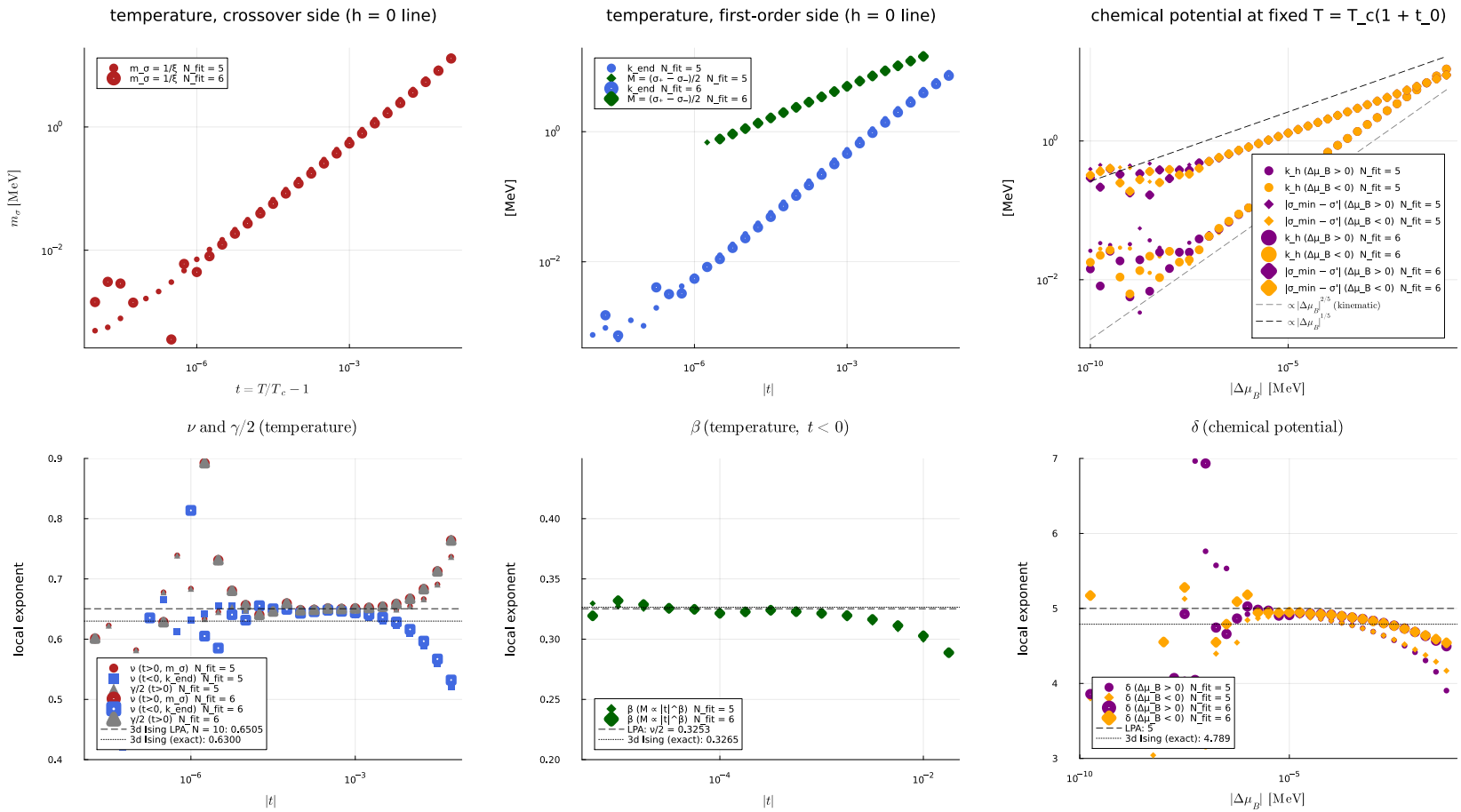


- Crossover side: $\xi \propto t^{-1/2}, \chi_\sigma = 1/m_\sigma^2 \propto t^{-1} \Rightarrow \nu = \gamma/2 = 1/2,\ \eta = 0$ ($\nu' = 1/2$ on both sides of the first-order line)
- First-order side: half jump $\propto (-t)^{1/2} \Rightarrow \beta = 1/2;\ T = T_c$: $|\sigma - \sigma_c| \propto |\mu_B - \mu_c|^{1/3} \Rightarrow \delta = 3$

Landau exponents $\nu = 1/2,\ \gamma = 1,\ \beta = 1/2,\ \delta = 3,\ \eta = 0$ vs. 3D Ising $0.630,\ 1.237,\ 0.326,\ 4.79,\ 0.036$

mean-field QM (PQM parameters, $L = \bar{L} = 1$), CEP at $(T, \mu_B) = (31.0,\ 878.9)$ MeV · **SY**, in preparation

QM model: LPA critical exponents at the CEP

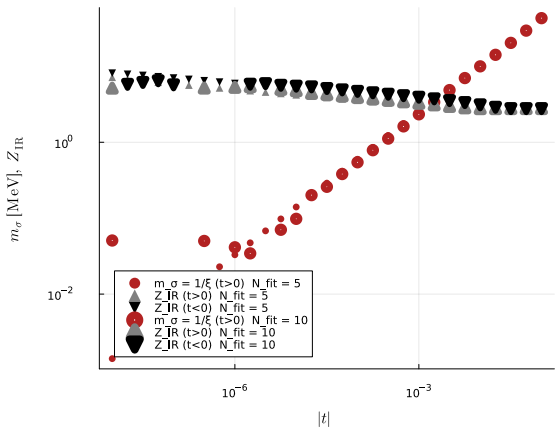


- Crossover side ($m_\sigma = 1/\xi$): $\nu = \gamma/2 \simeq 0.651$; first-order side: $\beta = \nu/2 \simeq 0.325$; fixed T: $\delta = 5$ ($\eta = 0$ in LPA)
- Plateaus over several decades of $|t|$: exponents agree with the LPA values for the 3D Ising class

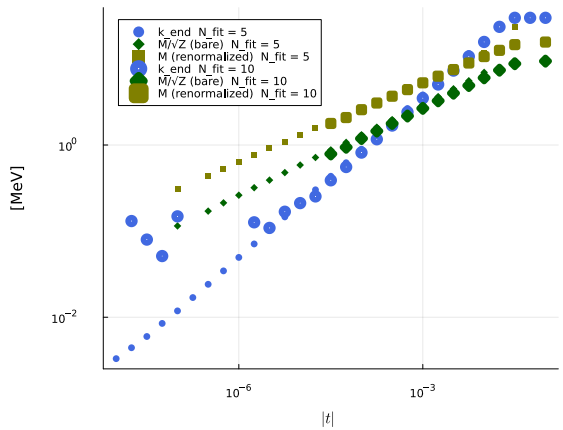
fRG LPA, QM model (no Polyakov loop), grid + Taylor ($N = 10$); CEP at $(T, \mu_B) \simeq (22.5,\ 857.1)$ MeV · **SY**, in preparation

QM model: LPA' critical exponents at the CEP

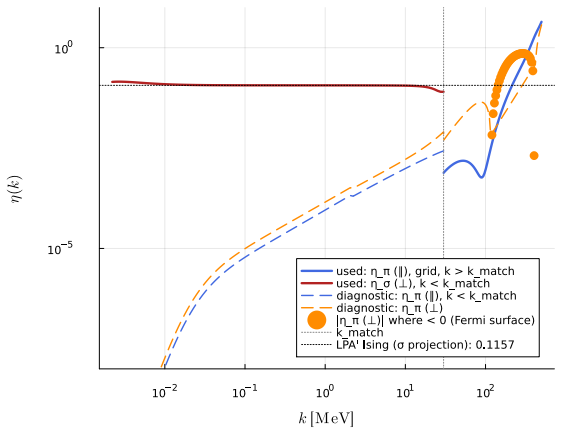
crossover side: m_σ; Z_IR both sides



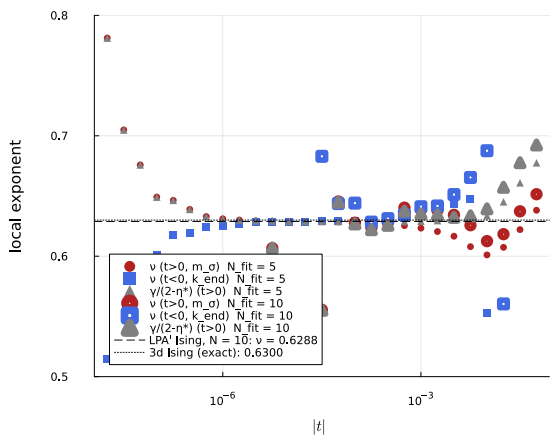
first-order side (h = 0 line)



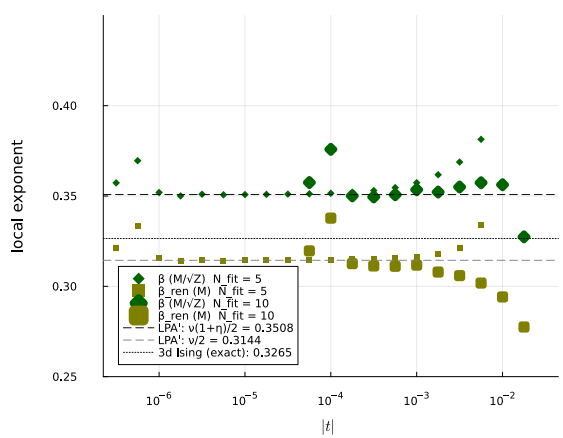
η along the flow at the CEP (N_fit = 5)



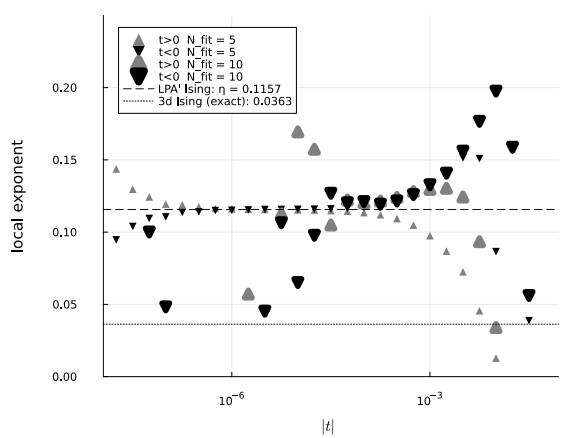
ν and $\gamma/(2-\eta^*)$



β (bare) and $\beta_{\rm ren}$



η from $Z_{\rm IR} \propto |t|^{-\nu\eta}$



- Anomalous dimension η from $Z_{\rm IR} \propto |t|^{-\nu\eta}$: $\eta \simeq 0.116 \neq 0$ (LPA: $\eta = 0$)
- $\nu \simeq 0.629$ (3D Ising: 0.630); $\beta = \nu(1+\eta)/2 \simeq 0.351$

fRG LPA' (σ-projected η), QM model (no Polyakov loop), $N = 10$; CEP at $(T, \mu_B) \simeq (37.1,\ 756.9)$ MeV · **SY**, in preparation

Critical exponents: MF vs. LPA vs. LPA'

	ν	β	η	δ
mean field	1/2	1/2	0	3
LPA ($N = 10$)	0.651	0.325	0	5
LPA' ($N = 10$)	0.629	0.351	0.116	4.38*
3D Ising	**0.630**	**0.326**	**0.036**	**4.79**

* from scaling relations

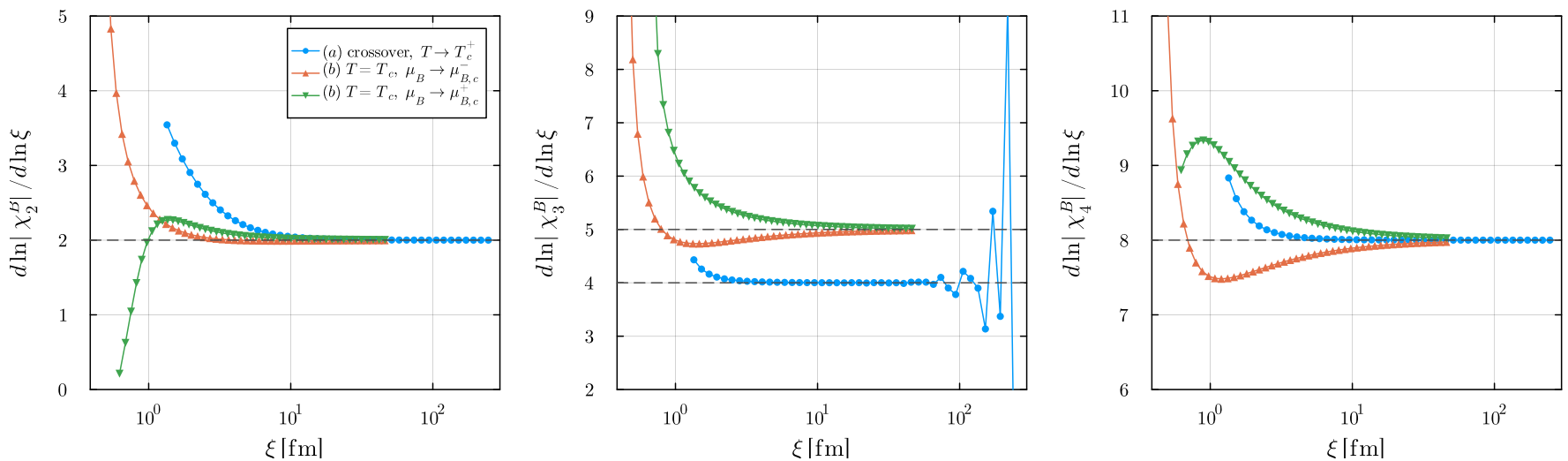
The field dependence of η is neglected in the present computation ⇒ exponents deviate from the 3D Ising class

The framework reproduces the expected hierarchy: mean field → LPA → LPA' approaching 3D Ising

SY, in preparation

PQM Mean field: exponents of baryon number fluctuations

$$\chi_n^B \propto \xi^{a_n}, \qquad a_n = \frac{d \ln |\chi_n^B|}{d \ln \xi} \ \text{(local exponent)}$$

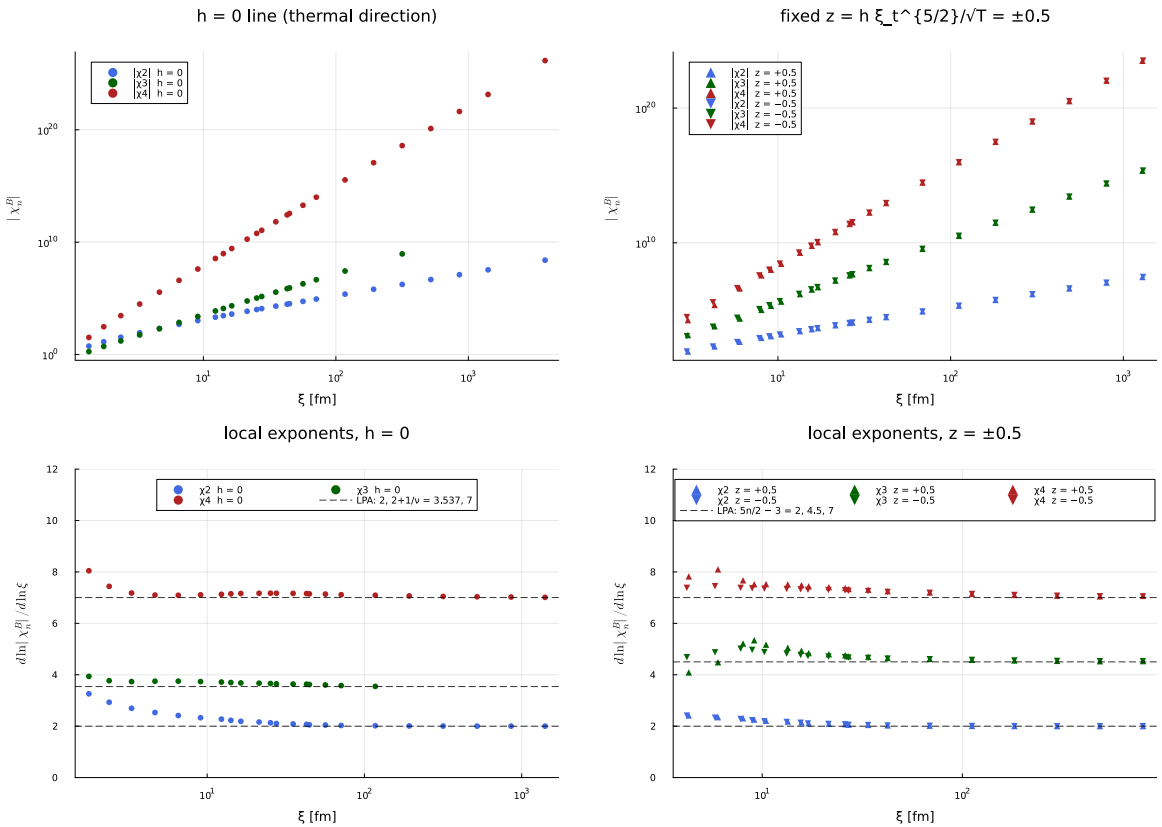


(a) crossover, $T \to T_c^+$ (b) $T = T_c, \mu_B \to \mu_{B,c}^{\pm}$

- Mean field: $\chi_2^B \propto \xi^2$, $\chi_3^B \propto \xi^4$ (a) / ξ^5 (b), $\chi_4^B \propto \xi^8$ vs. 3D Ising: ξ^2, $\xi^{4.5}$, ξ^7
- Asymptotic values reached only for $\xi \gtrsim \mathcal{O}(10)$ fm, beyond $\xi \lesssim 2$–3 fm in heavy-ion collisions

mean-field PQM · **SY**, in preparation

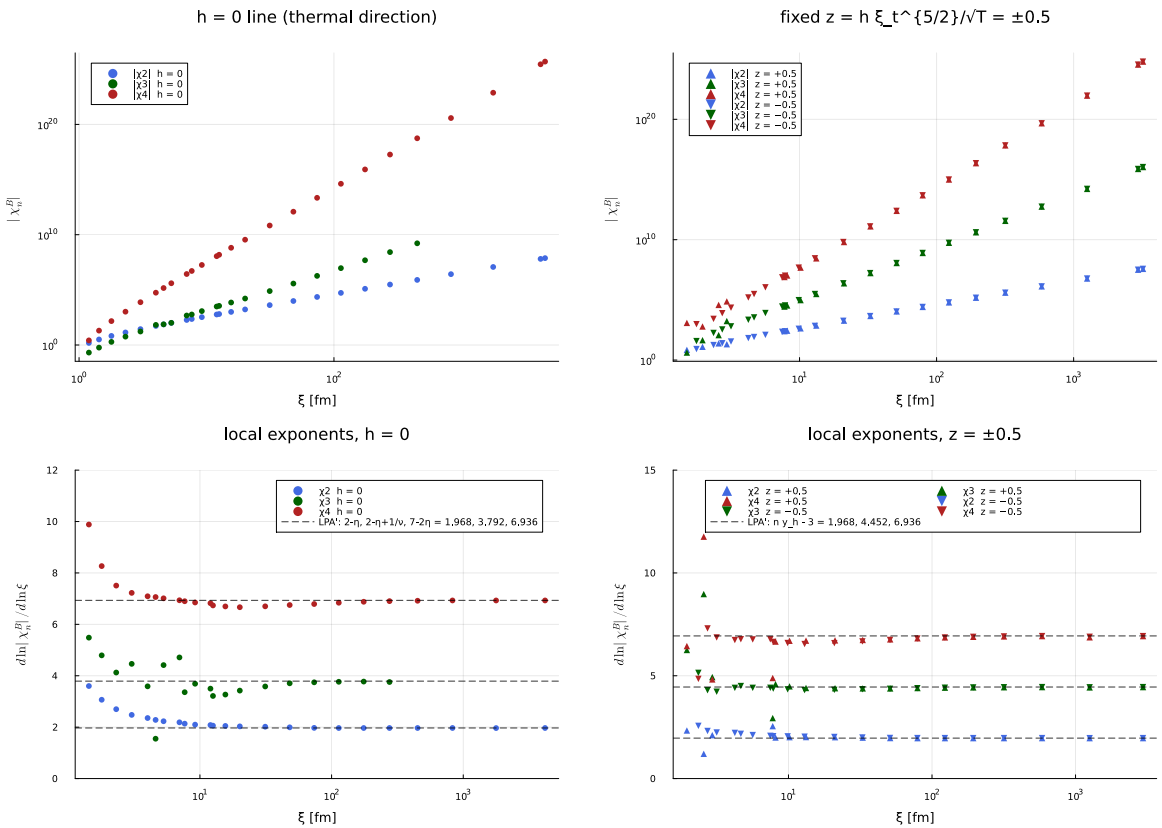
QM LPA: exponents of baryon fluctuations



- $h = 0$ (thermal direction): $\chi_2^B \propto \xi^2,\ \chi_3^B \propto \xi^{2+1/\nu} \simeq \xi^{3.54},\ \chi_4^B \propto \xi^7$
- Fixed $z = \pm 0.5$: $a_n = \frac{5n}{2} - 3 = 2,\ 4.5,\ 7$ ($\eta = 0$ in LPA)
- Clear plateaus up to $\xi \sim 10^3$ fm
- Compared to mean field ($\xi^2,\ \xi^4\ /\ \xi^5,\ \xi^8$): Ising-like scaling of χ_n^B

fRG LPA, QM model (no Polyakov loop) · **SY**, in preparation

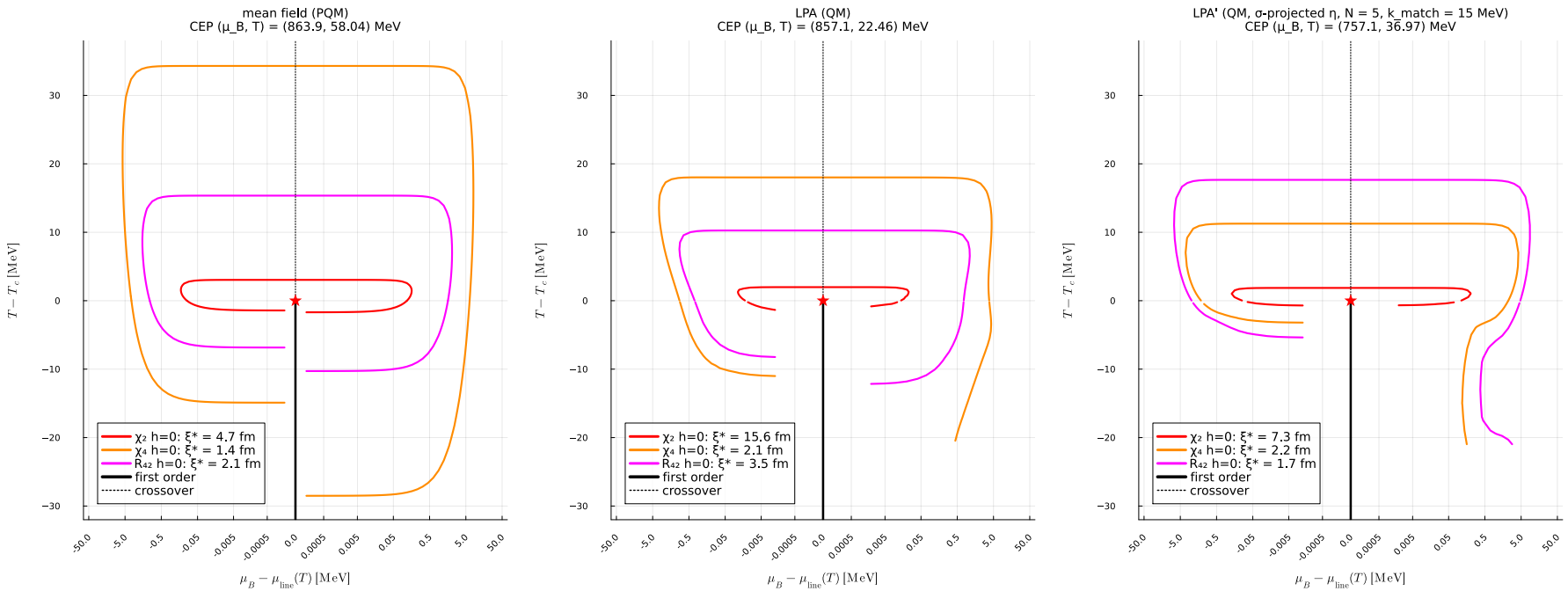
QM LPA': exponents of baryon fluctuations



- $\eta \neq 0$ shifts all exponents: $a_n = n(5-\eta)/2 - 3$
- $h = 0$: $2-\eta,\ 2-\eta+1/\nu,\ 7-2\eta$ $= 1.968,\ 3.792,\ 6.936$
- Fixed $z = \pm 0.5$: $1.968,\ 4.452,\ 6.936$
- LPA: $2,\ 4.5,\ 7$ → LPA': slightly reduced by η

fRG LPA' (σ-projected η, $N = 5$, $k_{\text{match}} = 15$ MeV, $\eta^* = 0.0318$), QM model · **SY**, in preparation

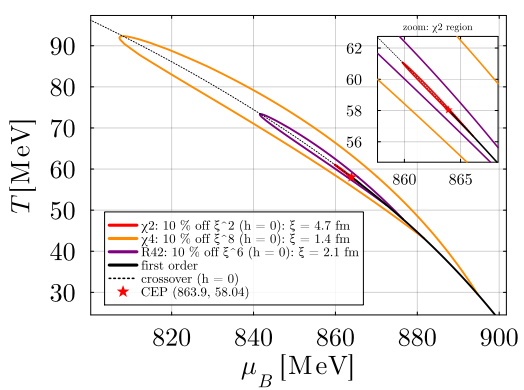
Critical region of baryon number fluctuations



- Near the CEP $\chi_n^B \propto \xi^{a_n}$; critical region: local exponents $d\ln|\chi_n^B|/d\ln\xi$ within 10% of their asymptotic values
- Tiny region: $|\mu_B - \mu_{\rm line}(T)| \lesssim \mathcal{O}(1)$ MeV; different for χ_2^B, χ_4^B, R_{42}^B and for each truncation

The freeze-out curve passes outside the critical region ⇒ experiments see the non-universal “ripples”

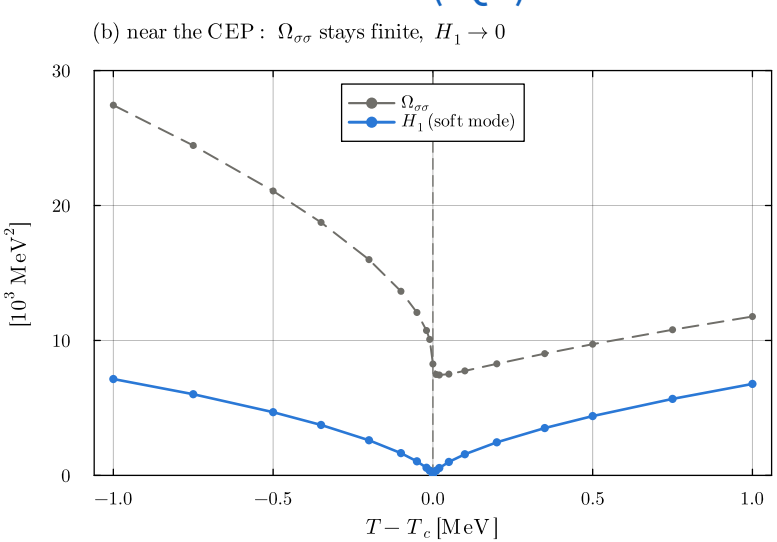
mean field: PQM; LPA, LPA': QM model · **SY**, in preparation



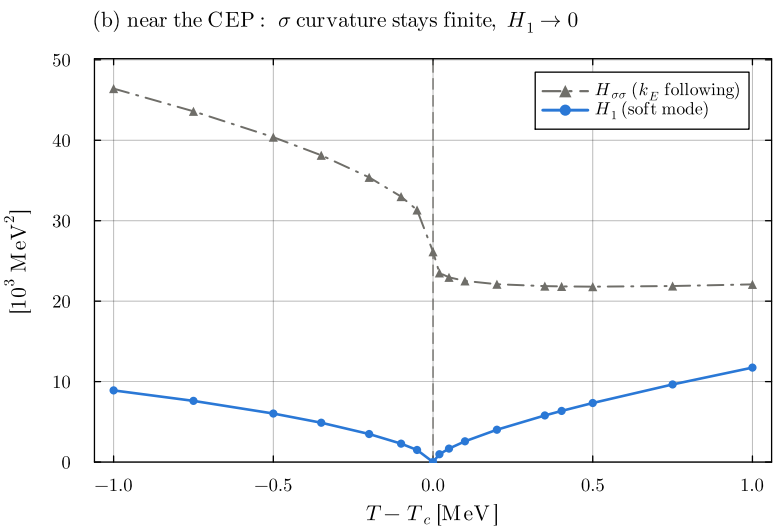
mean field (PQM): same regions in the (μ_B, T) plane

PQM: mixing of the Polyakov loop and σ

Mean field (PQM)



QCD-assisted LEFT (fRG)



through the CEP at fixed $\mu_B = \mu_c$

$$\Omega^{(2)} = \begin{pmatrix} \Omega_{\sigma\sigma} & \Omega_{\sigma a_3} & \Omega_{\sigma a_8} \\ \Omega_{a_3\sigma} & \Omega_{a_3 a_3} & \Omega_{a_3 a_8} \\ \Omega_{a_8\sigma} & \Omega_{a_8 a_3} & \Omega_{a_8 a_8} \end{pmatrix}$$

$$a_3 = gA_0^{(3)}, \quad a_8 = gA_0^{(8)}; \qquad \Omega_{\sigma a_3},\ \Omega_{\sigma a_8} \neq 0$$

$$H_1 = \text{lowest eigenvalue of } \Omega^{(2)} \xrightarrow{\text{CEP}} 0$$

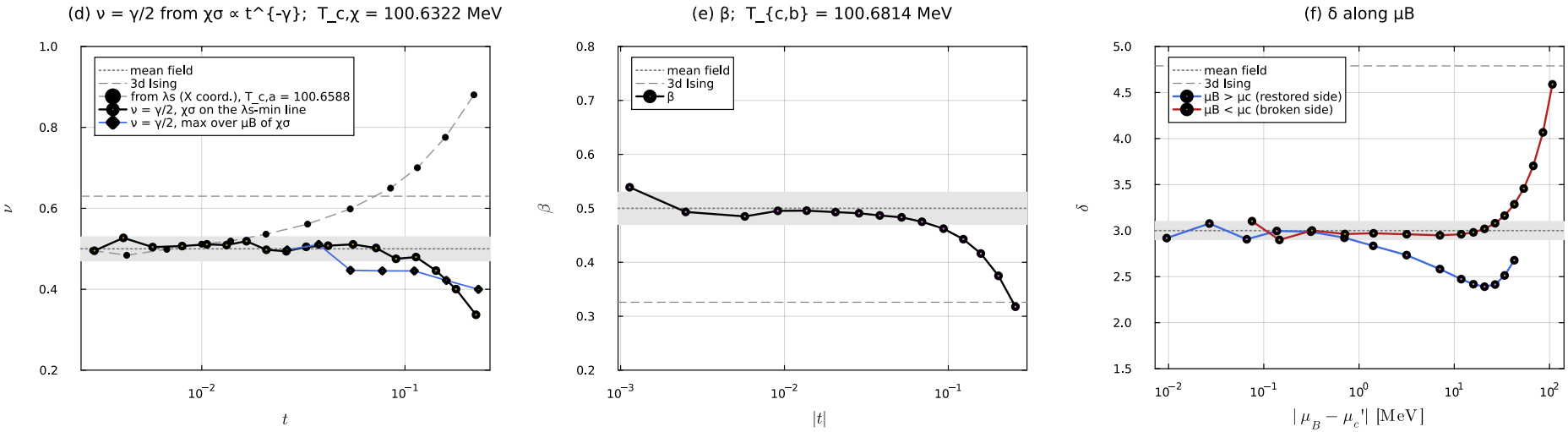
- σ couples to the Polyakov-loop variables (a_3, a_8) through the quark loop
- The soft mode H_1 is a **mixture** of σ, a_3 and a_8
- $\Omega_{\sigma\sigma} = m_\sigma^2$ stays finite at the CEP $\Rightarrow \sigma$ alone is **not** the critical mode

Haensch, Rennecke, von Smekal, arXiv:2308.16244

SY, in preparation

PQM at the CEP: mean-field behavior

QCD-assisted LEFT (fRG): local exponents at the CEP (grey bands: mean field)



- The Polyakov loop is an **input**: $V_{\text{glue}}(L, \bar{L})$ carries no fluctuations, no L propagator in the flow
- The soft mode contains an L component ⇒ part of the critical mode is treated at mean-field level
- ⇒ mean-field exponents at the CEP: $\nu = \gamma/2 = 1/2,\ \beta = 1/2,\ \delta = 3,\ \eta = 0$; QM model (no L): Ising-like

Outlook: a fluctuating Polyakov loop (A_0) is needed for 3D Ising behavior in PQM

SY, in preparation

Summary and Outlook

Summary

- The “ripples” of the CEP induce a **non-monotonic** behavior of baryon number fluctuations
- The nuclear liquid–gas transition may affect baryon number fluctuations in **low-energy** collisions
- The phase diagram at high density needs improvement: more degrees of freedom are required for a reasonable first-order transition line
- The mixing of the Polyakov loop and σ changes the **soft mode** at the CEP
- Fluctuations of the Polyakov loop must be included properly to obtain critical exponents **beyond mean field**

Outlook

- How strongly does the critical behavior of the CEP affect the fluctuations?
- Improve the LEFT code and compute fluctuations for systems with different CEP locations
- Use the resulting phenomenological data to study the relation between fluctuations, CEP location and freeze-out curve (Bayesian analysis / machine learning)